

Manga Collection Royale

Bachelor Thesis

Web Development

(23.10.2018 - 02.01.2019)

Janina-Cindy Ortelt

Development of a mobile application
for manga collectors

Always have an
overview of your
manga collection, no
matter where you
are and no matter
where you go.



Also YOU can get the "last man(ga) standing" in
your collection!

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TESTUSER: Janina

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TESTSITE FOR ADDING BOOKS: <https://bit.ly/2LAuBBI>

PASSWORD: 12345

API WEBSITE: <https://bit.ly/2Q1txao>

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PREFACE

This report focuses on the planning and development phase of a mobile application for scanning and managing books. In this case, the application specifically refers to the niche topic: **Manga**. The working title of the application is "**Manga Collection Royale**".

For the general research, analysis, planning and development of this bachelor thesis, methods and procedures were used, which have been learned during the study.

The report explains the approach as well as the detailed development and test phase of the application. It explains what things have been considered or less taken into account and why. All research methods and its results, as well as the main focus, is related to the **German market**.

In general, this project is a very large, which will be further developed after the bachelor thesis.



ABSTRACT

The goal of this bachelor thesis was to find out, if manga collectors permanently have an overview of their collections and how this could be improveable with the help of a mobile application.

It was supremely significant to develop a possibility, to simplify the scanning of the collection, since especially scanning of grand series like Detective Conan with about 95 volumes (conanwiki.org, 2018) can be incredibly tedious and time-consuming, if these volumes have to be scanned individually.

It was equally substantial to analyze other existing Apps and filter out their advantages and disadvantages.

The required information was carried out with self-research, surveys and explicit user tests, both with a high-fidelity prototype, as well as with the actually programmed prototype version of the Android app.

All results were included in the development phase and have contributed to the preliminary final test version.

As a result, both a visual prototype and a functional app were developed.



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INTRODUCTION

For many people, collecting books is not just a hobby, above all its a passion and so they are spending a lot of money for it. But what if passion becomes a problem?

This project is about a very specific area of books - manga. Manga is the Japanese word for comics. Comics from the Korean region, in turn, are referred to as Manhwa and comics from the Chinese area as Manhua. These names have been consolidated over time in the European area ([De.wikipedia.org](https://de.wikipedia.org), 2018).

In recent years, more and more manga from Japan have been licensed for the German market and the fan base is growing constantly. In addition to digitized manga, however, the printed manga particularly obtain their places in the German bookshelves.

Indeed minor events like a "reading weekend" or challenges named "read at least one manga every day" or "My Manga Year" have been launched.

Moreover, statistics about manga already collected are regularly published via YouTube or additional media to share it with other manga fans.

The current offered amounts of manga on the German market are enormous. Each month, various volumes come out on the market at separate times and from different publishers.

In the following, I will point out, which problems this passion can bring, the possibilities that exist and how they could be optimized or improved.

PROBLEM

For a long time, I am interested in collecting and reading manga. I also repeatedly inform myself about recent releases and get suggestions, inspirations and opinions of other manga collectors.

When buying used manga, I often find out that I can lose the overview of my already collected manga very quickly. Again and again I've to look into the shelf to check, if I already own a certain volume.

Similar problems can also be found of other manga collectors. A certain YouTuber, who regularly shares his latest purchases with his fanbase, indeed inadvertently regularly buys 2-3 copies of the same volume.

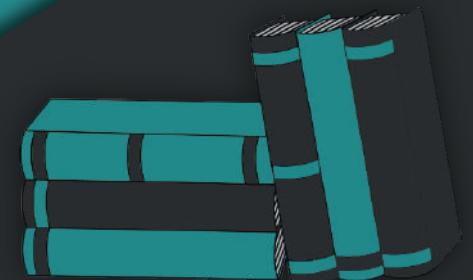
As more and more manga will be published on the German market every month, you can also lose the overview of them very often. Sure, you can make lists yourself, but many people do not have the time or desire for it ([Speciousness - YouTube](#), 2018).

Frequently you would also like to know which volumes are still missing in your collection, or which series are already completed.

Of course, some apps can make your life easier, but a lot of these apps have limited functionality. But with most of these apps, you have to scan each book individually, which can be extremely time-consuming for series with many volumes. For this reason, many people with a big manga collection do not want to invest their time for scanning and saving their books with a mobile app.

PROBLEMS TO SOLVE

- 1) Is the loss of the overview a general problem?
- 2) What information and functions are important?
- 3) What are the pros and cons of already existing apps?
- 4) How can I create an app that is adapted to the wishes and needs of other users?



ABOUT THE RESEARCH



(Rucker, 2016)

Research Question

Do people experience a general problem, to losing the overview of their collection and is it possible to develop a mobile app to support them, if the problem exists?

Is it possible to improve the scan system to scan books faster?

Research Purpose

Development of an app that caters to the needs and wishes of manga collectors. This app should be designed to enable a better scan function to capture and manage books faster. Above all, the app should provide the ability to store entire manga collections easier into the system, without having to scan each book individually.

It should additionally include substantial informations for a manga collector.

Research Objectives

Information must be gathered, on whether manga collectors actually have a problem that they definitely lose track of their collection. Further, it would be significant to know if they may already be using tools for maintaining their collections and if you are satisfied with them or not. What are the pros and cons of existing book management apps? Another important point would be to know what informations are important for manga gatherers.

The resulting results should be taken into account in the development phase of the app and optimized specifically by user tests.

METHODOLOGY

This project was developed corresponding to the Pragmatic Philosophy and is based on the abductive approach.

Qualitative as well as quantitative research strategies were chosen, resulting in a mixed method.

An Internet research was conducted to receive targeted and accurate information. Further, a general survey was made and accomplished of the target group. In addition, a competitive research was conducted to find out which products already exist related to this topic and what advantages and disadvantages they have.

Since the data had to be collected in a given time, the cross-sectional study was chosen.

Philosophy (Pragmatism)

For the project I chose the pragmatism philosophy. It allows me the objective as well as subjective consideration, which makes sense, since I by myself belong to my own target group.

In pragmatism, the world is viewed and interpreted in many various ways. You can't see from a minor capture view the whole picture, because there can be other truths.

Just because I recognize something as a problem, other people do not have to recognize it as a problem.

Approach (Abductive)

For the project I decided on the abductive approach. With this approach, I had the advantage that I did not have to explicitly refer to the inductive or deductive reasoning and in this way I gained more freedom in the research process.

With the abductive reasoning, a declared hypothesis is formed, which should ultimately expand the findings.

At the beginning, a hypothesis is made and then predictions based on the hypothesis are derived (deductive reasoning). Thereafter, facts are searched which should confirm the assumptions (inductive reasoning).

Before the start of the project, I have generally observed that many manga collectors, especially myself, have the problem of keeping the overview of the collection and that there are no dedicated apps on the market which fits to the needs of manga collectors.

The research should check more closely, if this ultimately undoubtedly represents a general problem.

Strategy (Internet Research, Competitive Research, Survey)

To gather the information, I needed, I chose 3 different strategies. To question the general situation of the problem, I searched the Internet for information. In doing so, I explicitly looked for platforms on which manga collectors like to present themselves.

For general information, I started a survey. At this place I wanted to filter out the common behavior and general opinion of the collectors.

Of course, I was already aware of people with smaller collections would probably have fewer problems keeping track of their books. Of course, I could not expect that all collectors who indeed have the same problem will also take part in this survey.

But it was innovative to discover how people principally address the topic, how they help themselves and what they would like to have.

The last method leads to a competitive Research. Of course, there are already apps with which it is possible to scan and manage books. It was supremely significant to discover the pros and cons and to develop a more accommodate app with the help of the results.

(Dudovskiy, 2018.)

User testing

Three different user tests were made in various development cycles.

During the first cycle, the first hi-fi prototype was created. This was extensively discussed with friends, acquaintances and relatives and suggestions for improvement were noticed.

In the second cycle, the hi-fi prototype was revised and improved on the basis of the suggestions for improvement. This prototype, in turn, has been tested by actual users. Based on a short interview with specific questions, the opinions of the users were recorded.

In the last cycle, the programmed app was tested. For this purpose, a person from the target group was selected who should test the function of the app. It had to be a person whose books already exist in the database.

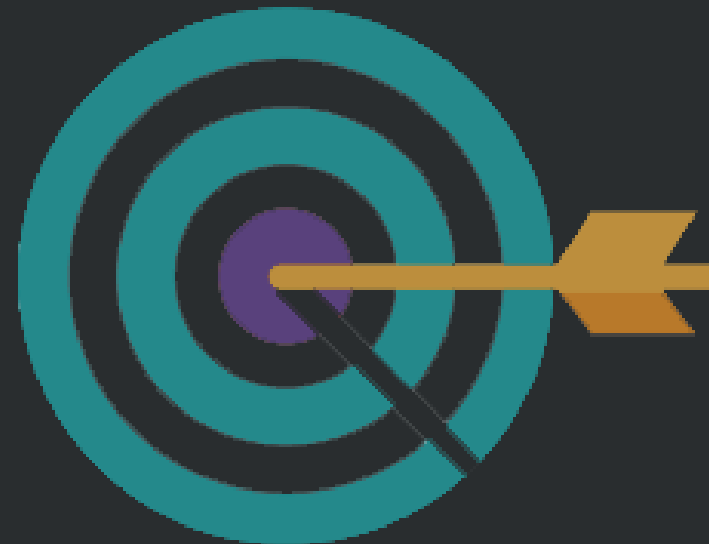
Time horizon (Cross-Sectional)

There has been a time limit of 10 weeks to complete this project, which means that I have chosen a cross-sectional study. This, in turn, provides only a small snapshot of the current situation.

A further poll and Internet research over a more extended period of time would possibly effect other results.

Methodological conclusion

With the help of this approach, I was able to enter the research with an idea. The individual strategies have returned various results, which can maybe relate to another basic precondition. In general, I could collect beneficial information that ultimately led to the result.





RESEARCH PROCESS

Beforehand, I want to explain, what has motivated me to address this topic. I want to show which informations have been collected in advance and how I finally came to the result of the actual idea.

I by myself have a medium-sized collection of manga and sometimes have the feeling to lose the overview of it. It also annoys me every time when buying new or used books to constantly have to look at my shelf, to check if I already own a book.

Especially at events like „Manga Conventions“, where you can also purchase books, you don't constantly have an overview of the already existing manga in your shelf at home.

For this reason, I thought about how could I create a practical overview.

Notebook:

One variant could be to write down all purchased manga fine and neat into a notebook. But what if you want to sell books again or don't exactly know how many volumes belong to a series? Even a notebook can become confusing very quickly someday.

Excel:

Another practical option could be to create an Excel spreadsheet. However, it is also very time-consuming to enter everything by hand. Undoubtedly, with a few macros and functions, things can be facilitated, but this is rather impossible for people without the corresponding knowledge.

Mobile App:

Another variant, and probably the most comfortable in the modern age, is, of course, a mobile app. After all, the smartphone is always with you wherever you go. But everyone knows the range of apps is always enormous and not every app is the way you would like it to be. Each app has its advantages and disadvantages. In the following I will describe my research approach.

1. Internet research

At first I tried to find out if the problem of losing the overview about the manga collection could be a general problem. For example, how often does it happen that volumes are bought twice? It was also important to know what information and functions might be important to people.

2. Questionnaire/Survey

In the next step, I wanted to ask people for information and opinions through targeted questions. Are they already using apps and how satisfied are they with them? Which functions are most significant to them and how important the design would be?

3. Competition testing

In this step, I have looked at the current range of apps on the market that can be found for the topic and evaluated its advantages and disadvantages.

4. Prototype testing

The research and development phase additionally include a testing of the prototype, which I will explain in the Development section.

Internet Research

1. Step

Of course, researching a niche topic on the internet is not incredibly easy. But when you represent a part of the scene, you know exactly where to look. In a general sense you will gain frequently on information via Facebook, Twitter or YouTube.

Ever since I started to get interested in this topic, I've seen over and over again, that some people have purchased the same books twice by mistake. That's because they don't have an overview of their own collection.

Therefore, it was possible for me, through targeted searches, to come across some of these cases. There, books that were bought twice are offered for exchange or sale.

As a conclusion you can see it can be a widespread problem. Both that people apparently can lose track of their collection, as well as that they inadvertently buy the same books twice.



search result on facebook
(www.facebook.com, 2018)



search result on Twitter
(Twitter.com, 2018)

The second part of this research was to find out what information people share with others and what information are generally important for them.

Through personal discussions and own interest, I figured out, that it is important to know which volumes will be published within the actual or upcoming month. Information about the status of a series and when the next volume appears, seems to be very important, too.

To gain a thorough overview, it is of course equally significant to know which volumes of a series you already own.

Moreover, manga collectors seem to like to share statistics about the numbers of collected and read manga. They also like to publish information about their newly purchased manga of the month.

Wish lists and shopping lists also seem to be rarely uninteresting. Since many people sometimes lend their manga to their friends or family, an overview or list about it might be appropriate.

Also, as already experienced, they like to sell or trade double or no longer used manga.



(JannySparrow, 2018)
twitter profile



(MelsMangaWorld, 2018)
twitter profile



(giira_ookami@Otakulife, 2018)
twitter profile

Survey

2. Step

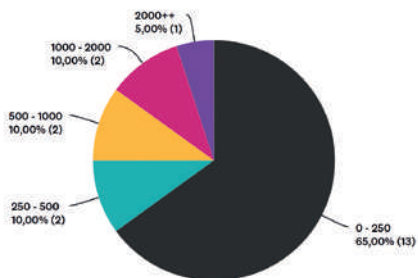
For more detailed information, I've created a survey and shared it with online platforms like Twitter, Instagram, or Discord.

All in all, unfortunately, only 20 people participated in this survey. Therefore, I will not rate the result of this survey very high, but will nevertheless consider it.

1. How many manga do you own at the moment?

Wieviele Manga besitzt Du zur Zeit?

Beantwortet: 20 Übersprungen: 0



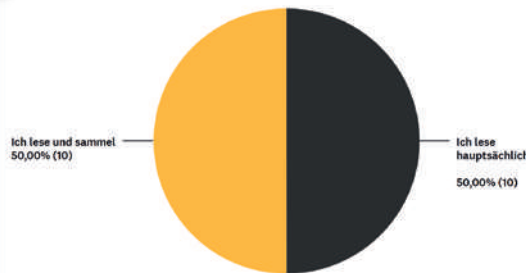
It was important to know how big the other people's collections already are. In this place you can see that the collections on the shelves of participants are substantially smaller.

This can have several reasons, like lack of space. But the costs can equally be an issue here, because the cost of the books is not low. For students it can be a certainly expensive hobby.

2. Do you collect manga or are you just reading?

Sammelst Du Manga oder liest Du Manga?

Beantwortet: 20 Übersprungen: 0



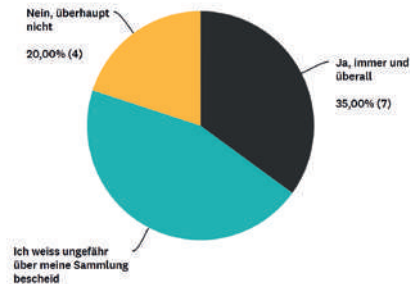
In the next question, you can clearly see the blocks are divided. Half of the people does not just read manga, they also collect them, but the other half only likes to read.

This probably also reflects the lack of space in the shelf or the cost issue again. Some manga fans prefer buying their manga digitally or lend it. Of course, there are also platforms that provide Manga for free, which naturally isn't always legal.

3. Do you have an overview of your manga?

Hast Du eine Übersicht über Deine Manga?

Beantwortet: 20 Übersprungen: 0



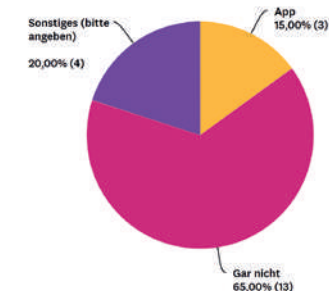
One of the key questions was to find out if people still have an overview of their collection, or have they have already lost it.

At this moment only a small part still has a comprehensive overview of their collection. Which in turn suggests that this could refer to the smaller collections.

4. How do you keep the overview of your manga?

Wie hältst Du die Übersicht über Deine Manga?

Beantwortet: 20 Übersprungen: 0



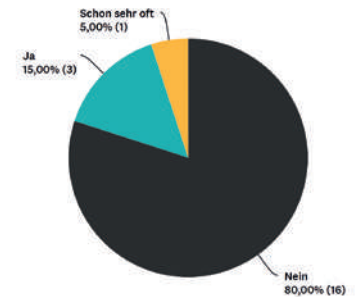
It seems, that most collectors currently are not interested or conceivably have no time or desire to gain an overview.

Maybe their collections are not that big, so that they don't need to get an overview. Maybe it is too cumbersome and time-consuming for them.

5. Have you ever bought a manga twice?

Hast Du schon einmal versehentlich Manga doppelt gekauft?

Beantwortet: 20 Übersprungen: 0

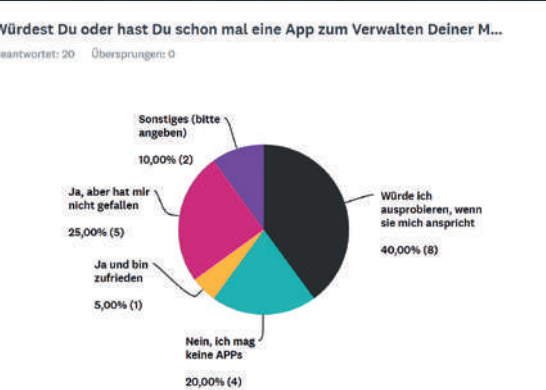


For the surveyed manga collectors, it hasn't happened frequently that they bought a manga twice.

After all, most of them have a small collection and so they still have a good overview of what they already have and what they do not.

Nevertheless, with a more comprehensive collections, the inadvertently double shopping of a manga can represent a problem.

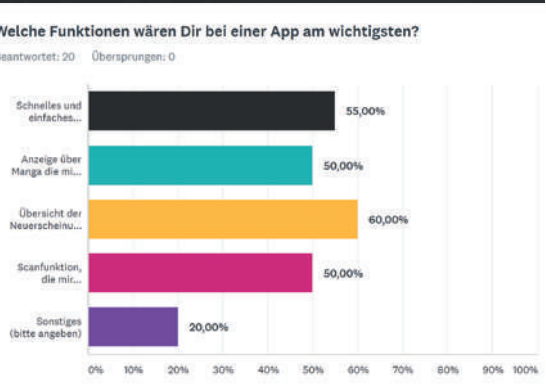
6. Would you or have you ever tried an app to manage your manga?



A lot of people are very open-minded when it comes to using an app. Some have already tried apps, but were obviously unhappy with them and may have stopped using them.

Only one of the respondents is satisfied with the choice of his app. A substantial part would be at least willing to use a good and attractive app if it convinces them.

7. Which functions are most important for you?

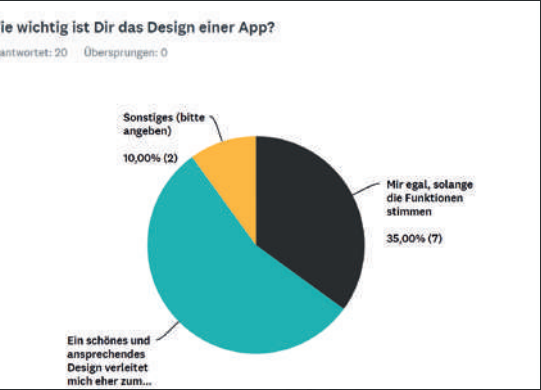


But which functions are significant now? Here, the four suggested functions all seem to be equally important. However, the monthly overview of recent releases, followed by quick and easy scanning of the books are relevant.

An overview about mangas, which are missing in a series, should be included. Also a scan function, which shows, if you already have a manga should be considered.

As additional functions a Note/Favorites list was desired, on which you can set the mangas, which you would like to buy next. Also, an orderly overview of existing and read manga was desired.

8. How important is the design of an APP?



Lastly, I wanted to know how important the design of an app could be.

Surprisingly, many people think the design is not so unimportant. A outstanding and attractive design seem to be one reason to install an app.

But on the other hand, for some people, the primary thing is that the functions fit.



Competition testing

3. Step

1. Meine Bibliothek

(Keith, 2018)

In order to develop something good and unique, you should also pay attention to your competitors. There are always things that have already been implemented well or less well.

So it can serve as an advantage to improve already well implemented features with your own ideas and expand or rebuild it. (Levy, 2018)

This app offers both the option to save books and comics. You can scan your books with an ISBN scanner, and you also get information about whether you already own a volume.

All scanned volumes are listed by name. But you also have the possibility to sort them by author. You can also see the results by categories (genres), series title or filter by comments. You also have the opportunity to create a wish list.

In general, this app is a really good solution, but you often have to add some information by yourself or customize the title, otherwise the books are displayed in a different order.

For a normal book collection, this app is a really good solution, but for a manga collector there are missing some useful information.

In addition, for a manga collection, it would be an advantage to list the series titles first.

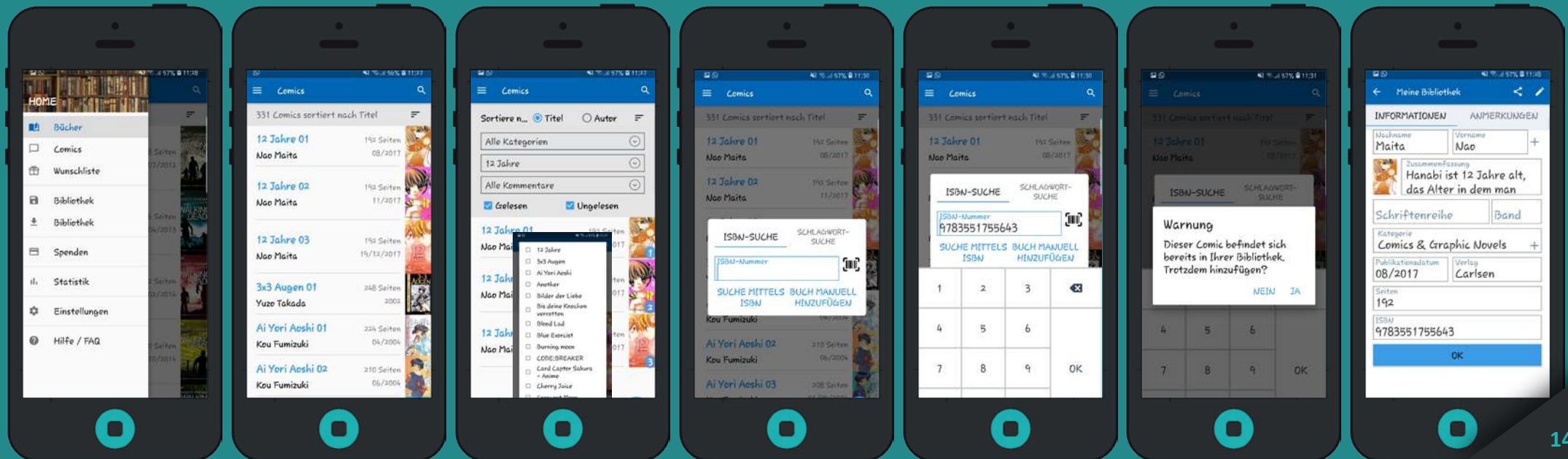
Because if you have several series with a sufficient number of volumes, you have to scroll very long. Sure you can use filters, but you equally have to scroll down, if the list is extensive.

Pros

- * scan function
- * wishlist
- * simple overview
- * warning, if a book already exists
- * Filter function

Cons

- * long scrolling if many volumes / series
- * often have to add information by yourself
- * not a nice design
- * long scan time (3 steps)
- * Each book must be scanned one by one



2. Bibliothek

(Rebensburg Solutions, 2018)

Another tested app, which presents some advantages, but also many disadvantages.

The listing of the manga is kept very simple, which brings a clear overview. However, all individual titles of a series are listed from top to bottom. For filtering, you can use the search field.

Of course, this app additionally offers a scan function. However, this can be really annoying in this app. You have to take 3 steps alone to scan a book at all.

The books are usually recognized, but also here you often have to adjust information.

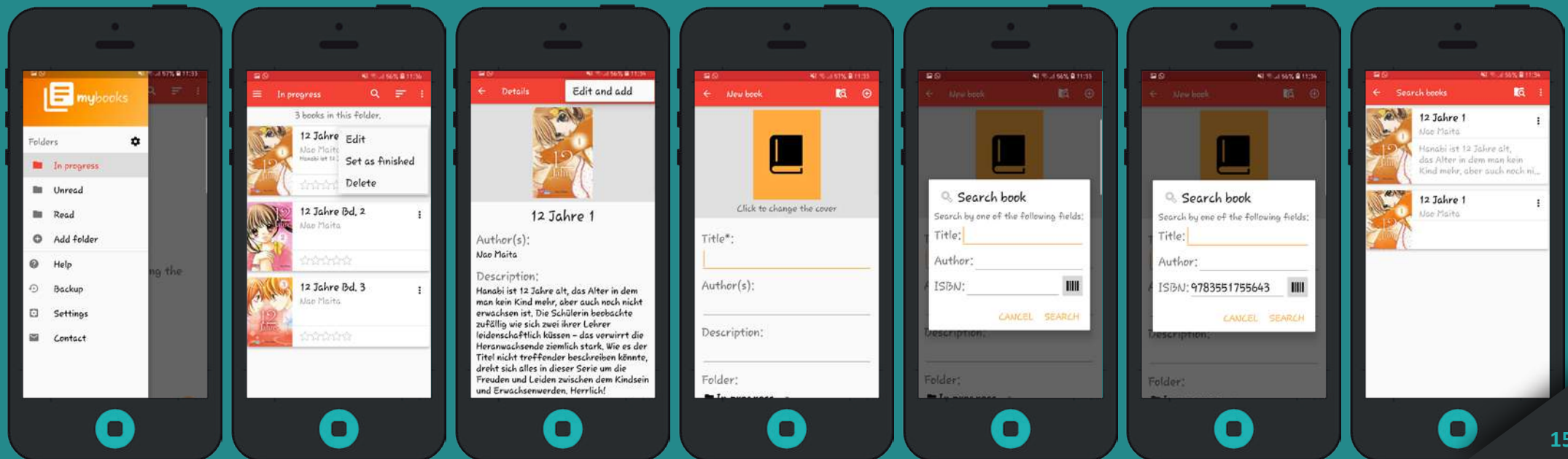
An attractive idea is the possibility to create your own folders. This allows you to categorize your books into different categories.

Pros

- * scan function
- * individual folders
- * simple clear overview

Cons

- * annoying long scanning
- * boring design
- * Some Infos have to be entered by yourself
- * no series overview



Conclusion

From a certain size of a collection, the loss of your own overview is seemingly inevitable.

If you have 100 different manga series and therefore 1,000 manga volumes, you can't always remember which series and volumes you already have or not.

Especially if you mostly buy second hand goods. Because especially the hunt for already out of print manga, can be very difficult and then it would be annoying, if you don't exactly know which volumes you already own.

By researching it, I could at least realize that it is not a huge dilemma, but it can be a minor problem for many collectors. And the larger the collections, the bigger the problem with the overview.

The current apps, which are on the market are more specialized for books in general and not for manga or comics.

Although they offer a scan function, but often this is very cumbersome, so it takes a long time to save a book at all. Principally with large manga collections, these functions can be absolutely time draining.

Because of this reason, many collectors have not ventured to these apps.

During my own app test, I figured out, that none of the chosen apps had a good and informative overview. All manga were merely listed with a name or author.

But there was no specific information about the series, such as when the next volume appears or if a series is already finished.

There are just overmuch important information and features that a manga collector could need.

So I came to the decision to develop an app for exactly this target group with all these significant informations and functions.

DEVELOPMENT APPROACH

Idea



Since I have recognized, that there is currently no good app on the market, especially for scanning and managing manga and comics, I have decided to design and develop an app that is specifically adjusted to the desires and needs of this target group.

For that, I would like to develop a different design, which has a certain recognition value and is linked to the manga/anime theme. The design should be appealing and arouse people's interest.

On the subject of information and functions, there is a very long list, which will not be uninteresting for the target group. Generally, this app is going to be a very large project and should be adapted to as many wishes and needs as possible so that many people will see a benefit from it.

Features that this app could/should include:

A general overview of the manga series inclusive detail view with information on the number of volumes, status of the series (completed, next volume) possibly highlighting which volumes you already possess from this series. A search function should also be integrated. In addition you should have further filter functions like, Author or next volume, etc.

An overview of the already collected series, with brief information on how many volumes you already own. Additional filter function so that only completely collected or incomplete series can be displayed. Of course, there should in addition be a search function and other useful filter functions.

Both in the main view and in the series detail view, there should be a small statistic (how many series, how many volumes, how many are read or unread, etc.)

In the series detail view, it should also be highlighted which volumes you own and which are missing. You could on top add a filter function (read, unread, existing or missing volumes).

One of the most key features, of course, is the scan function. It should be possible to scan books quickly and easily, so that you do not need too much time for scanning. A book should be scanned and stored directly in the database or removed. In this way, you should also be able to save a book as read.

Another addition should be the information scanning, with which it should be possible to know whether you already own a volume or not, or whether you have read it or not.

To simplify the scanning process of large series, it should be possible to scan just one book of a series. After successful scanning you can select other volumes and save or remove them.

An overview of the individual publishers or about the authors would probably not be uninteresting. In addition, especially a wish list/shopping list was desired or an overview with the monthly recent releases. It could also be added more statistics or a ranking.

The app could be extended with many features, which will make it to a really comprehensive app. Of course, not everything can be realized in the short project phase. That's why I'll focus on only a minor part of the functions.

Current Function ideas:

- General overview of the manga (series/volumes)
- **Overview of your own manga (series/volumes)**
- Filter functions
- **Scan functions**
- **Login/Registration**
- Wishlist/Shopping List
- Recent releases
- Publisher overview
- Author overview
- Exchange and sell
- Statistics
- Rankings



1. Phase Planning & Preparation

Expiration of the 1. Phase

Before I could go to the actual programming phase, certain things had to be prepared and planned. How should the system work, which functions should it contain, how should it be designed and how should it be realized in the end? As a result, the following planning and preparation has been worked out.

Interface Design and Heuristics

Sketch

First of all, I decided to follow Jakob Nielsen's 10 Principles of Interaction Design (Nielsen, 1994).

These guidelines are very easy to understand and prevent many problems during and after development.

Therefore, it is certainly a great advantage to follow these guidelines. You should always keep in mind who you are developing the system for.

Information Architecture (sitemap)

In order to develop an overview and an idea of how the system should work, I have created an information architecture sitemap. It will also show, how to navigate through the app. The further development process has been built on it, which made it much

The first idea of how the app could work was sketched on a paper. Just the actual idea was considered, which can later be rebuilt or expanded.

MockUp / Wireframe

In the following step a small & clear mockup was created based on the sketch. This will indicate the functions a bit more accurately.

User Flow Chart

With the User Flow Chart, I wanted to capture a more attentive look at how the system should operate. What should be spent or performed under which condition? This is essential for giving feedback on specific actions.

Prototyping

1. Cycle

In the first cycle, a first version of the hi-fi prototype design was created. This prototype was tested by teachers, known relatives, and myself and were referred to the guidelines. Ideas for improvement were worked out through feedback and evaluation of the test results.

2. Cycle

In the second cycle, improvements were made to the first example design. Afterwards, an interactive prototype was created from the prototype, which was then tested by users from the corresponding target group. The feedback was accordingly collected and analyzed.

Database

In the final part of the preparation phase, the possibility of data collection was considered. But how should the data be stored and managed?

To answer this question, I decided to develop my own database for the system.

In the planning phase, I have worked out an ER model for the database. However, this can be revised and expanded with additional functions and improvements.

Interface Design and Heuristics

Interface design represents the visual design of an interface through which the user can interact or communicate with a machine or system. Through inputs, whether by voice, facial expressions/gestures or by interaction, the user can "trigger" functions that tell the device what action to take.

This is typically designed with corresponding user interfaces like for example action buttons.

The interface design should therefore be tailored to the potential user. Therefore, it has to be designed and programmed as user-friendly as possible, clear & easy to understand and good and easy to use. It should also be self-explanatory to the user and as effective as possible.

It must be uncomplicated and not confusing, otherwise the user will stop using it. Furthermore, a user interface should not be overloaded with functions and interactions, as otherwise it may "crush" the user. Sometimes less is more.

(Gruenderszene.de, 2018)

For the development of a good interface design there are certain Interface Design Guidelines, which are also called heuristics. The most well-known and widely used guidelines were developed by Jakob Nielsen together with Rolf Molich in 1990, which Nielsen revised in 1994 and which are still used today. (Meiert, 2005)

1. Visibility of system status

For a good user experience, feedback is extremely important for the user. The user should always receive feedback on both successful and error actions.

2. Match between system and the real world

The system should use the language the user knows from the real world, and which matches the corresponding user group. For example, an app for doctors should also contain texts written in the appropriate vernacular.

3. User control and freedom

It can happen that a user performs actions that were not intended. Therefore, it must always be possible to undo such actions or perform them again.

4. Consistency and standards

Humans are creatures of habit and love to reuse them. That's why you should not reinvent things that have proven to be good. Standards help users to intuitively navigate, which helps the user to get along with the interface.

5. Error preventing

In order to avoid unnecessary expenditure of error messages, it is important to design and program the system so that the user can make less mistakes.

6. Recognition rather than recall

With a lot of information, it is impossible for the user to remember or perceive everything. Therefore, it is necessary to highlight the most important information, or keep information as simple and clear as possible.

7. Flexibility and efficiency of use

The user should always be able to perform actions flexibly. There should not be just one way to land on specific actions.

8. Aesthetic and minimalist design

Communication always includes text, colors, buttons, symbols and pictures that a user can perceive. In order not to overfill a user with such information, you should keep the use of these communication methods as low as possible. Else it can happen that the user won't recognize the important and relevant content.

9. Help users recognize, diagnose, and recover from errors

Also error messages should be kept as simple as possible. These should tell the user what the error has caused and suggest how to fix it.

10. Help and documentation

Some actions or mistakes are not always self-explanatory as they may be new. Therefore, you should give the users a detailed help. For example, new features could be explained with little tutorials, or you could add a FAQ section where the details are explained.

There are many benefits of using and maintaining these guidelines, which for example will make designing and programming much easier.

Reducing errors by users

If the user understands the interface and its actions, and the user is passed through the system with appropriate feedback, they will automatically have fewer opportunities to make mistakes.

Less frustration

Of course less mistakes mean less frustration. If a system is designed to be too complicated, so that the user can not understand the navigations and actions, it can also be very frustrating.

Increased confidence

If the user is satisfied with a system and easily gets the information he desires, it also increases confidence.

Increase satisfaction

When the system is designed and adapted to the user, it also increases satisfaction. Which ultimately ensures that products will continue to be used.

Increase reuse rate

People who are satisfied with something tend to use things again and again because they know what to expect and are satisfied with it.

Reduce development cost and time

Errors that can already be avoided during the development phase can also reduce the costs and time of development.

These aspects and guidelines I will take into account in my planning and development phase to the best of my ability and to find out through appropriate tests, how well I could finally implement them.

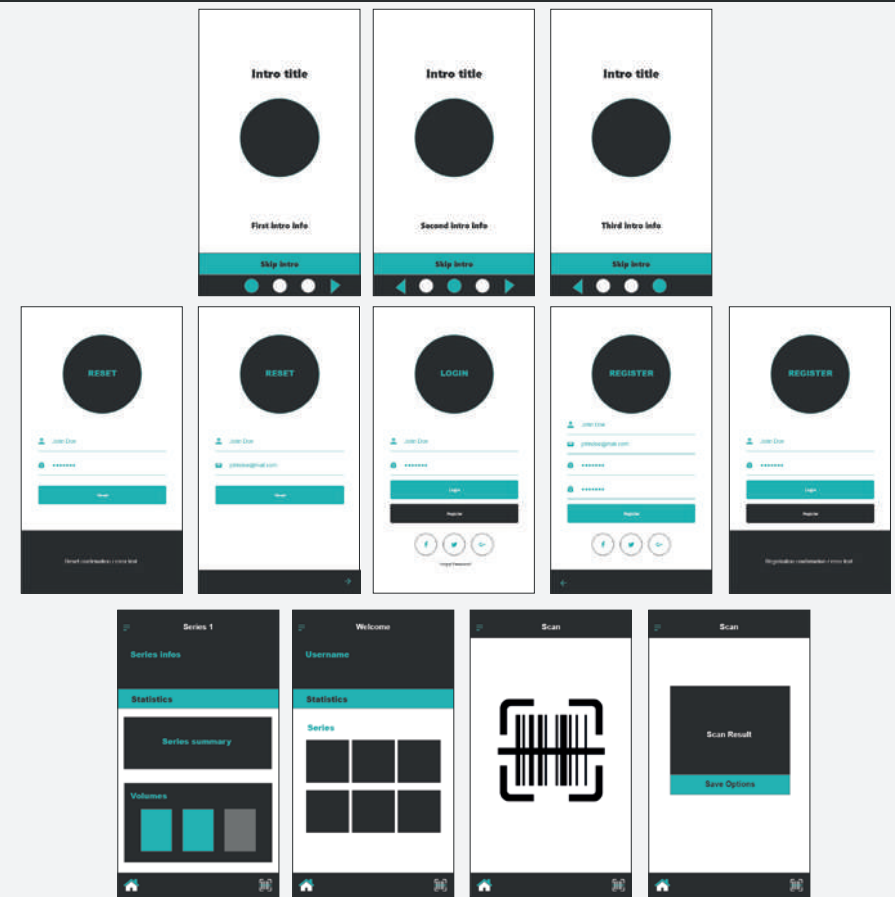
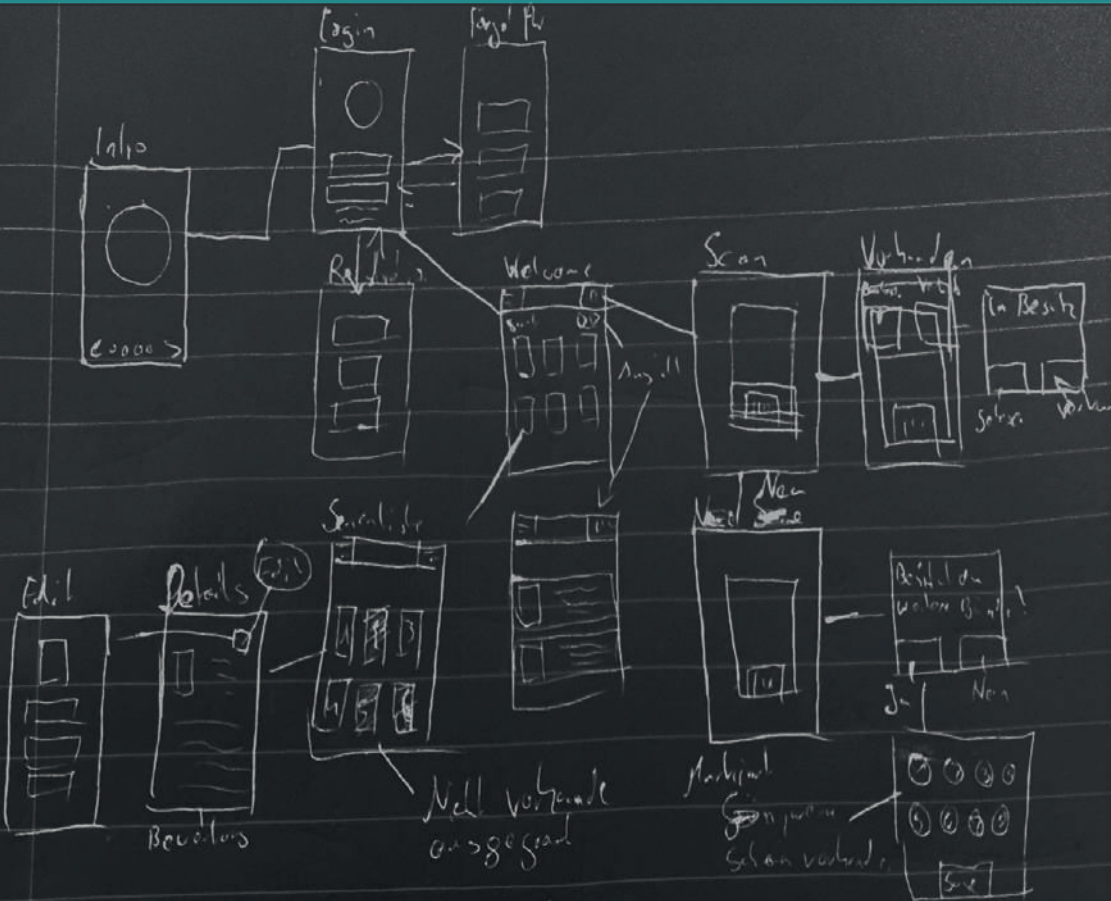
(Meiert, 2005) & (Nielsen, 1994)



To design a website or app, you first require a good thought out information architecture which should be understandable by the potential user. It is therefore necessary to design it as user-friendly as possible and less complicated, because with the information architecture you should guide the user the way through the app. (Pikover, n.d.)

Information Architecture (Sitemap)





After the first thoughts about the functionalities and navigation of the app the first ideas were worked out and a sketch were created.

Based on that, I've designed a first small visual Mockup, which should show the navigation and functions more exactly.

The app could start with a little introduction which promotes

and highlights the key features of the app. Of course you should be able to skip it, too.

After the Intro, the login process follows. Here of course you have the possibility to log in. If you don't have an account, you also have the possibility to register. You can also reset your password, if you've lost it.

After you have successfully logged in, you will be redirected to the main screen.

At this place you will find your already scanned series and a small statistics about the number of series, volumes and the number of read and unread manga.

If you "click" on a series, you will be guided to the detailed view.

On this page you will find a small summary about the series and an overview about the individual volumes.

Volumes, which you already own should be clearly highlighted. Volumes, which you are missing, should be grayed out. Once more, a small statistic should be displayed.

In addition, further details about the series could be

Both from the main menu as well as from the detail view, you should have the possibility to reach the scan page at any time.

With the scan a small UI appears, with which you will have access to further functions.

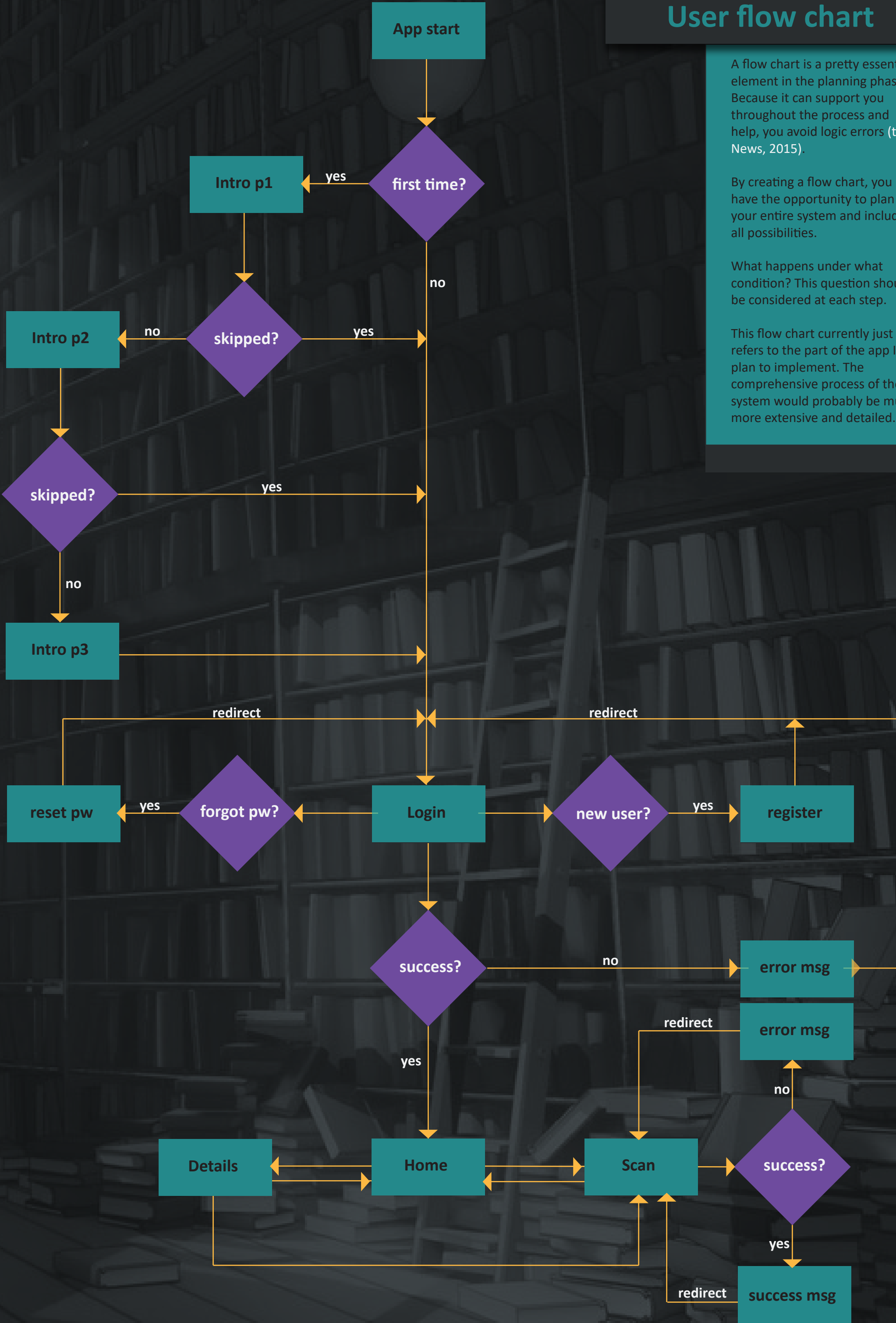
User flow chart

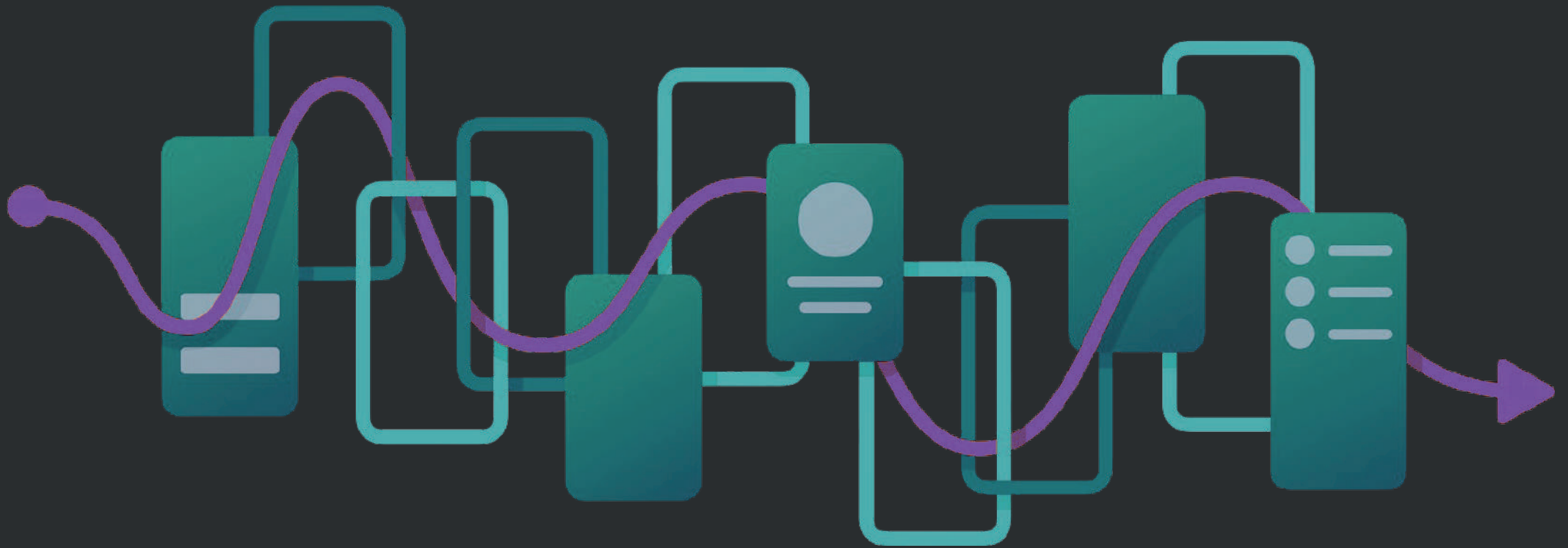
A flow chart is a pretty essential element in the planning phase. Because it can support you throughout the process and help, you avoid logic errors (t3n News, 2015).

By creating a flow chart, you have the opportunity to plan your entire system and include all possibilities.

What happens under what condition? This question should be considered at each step.

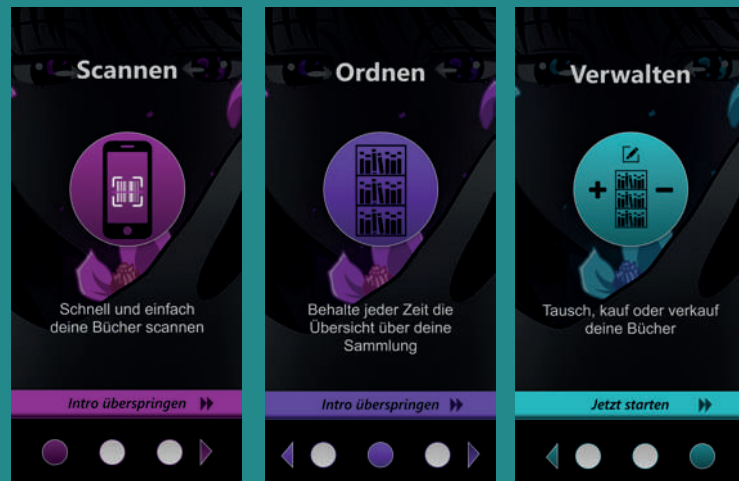
This flow chart currently just refers to the part of the app I plan to implement. The comprehensive process of the system would probably be much more extensive and detailed.



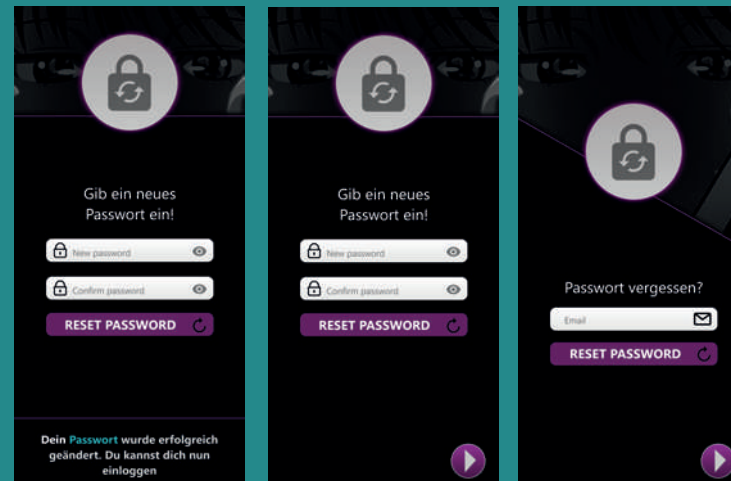


PROTOTYPING

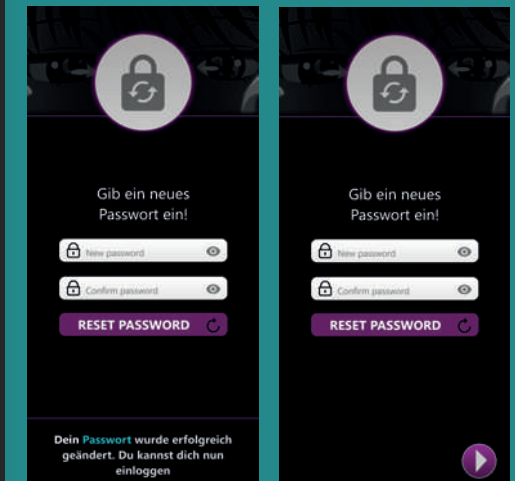
INTRO



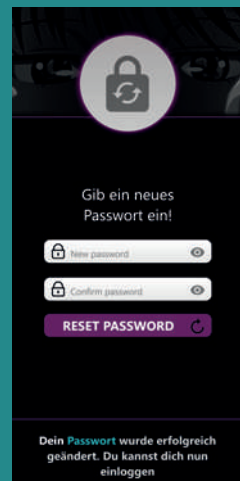
PASSWORD RESET



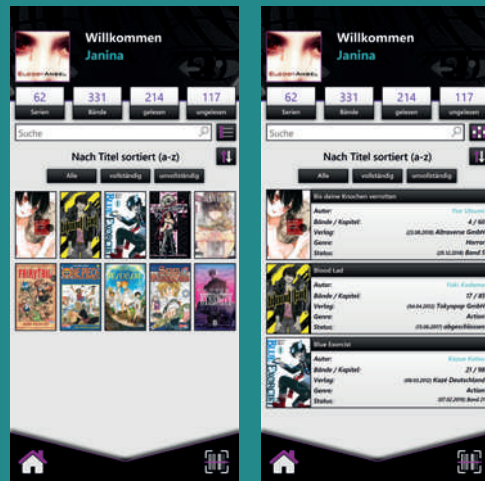
REGISTER



LOGIN



WELCOME



DETAILS & FILTER



Optimization

Based on the feedback and inspirations from other apps, I've reworked the design.

My goal was nevertheless to pay attention to the design, that it fits the actual topic and still sets itself apart from other standard designs.

The overview on the welcome screen has been designed a bit clearer and better recognizable. Nevertheless, I have tried to focus more on a app related design.

Especially in the list view, I reduced everything to the most important information and suggest more emphasis on optimal view so that the information can be better perceived by the user.

I have adapted the "detail view" to the clearer design and highlighted important information accordingly. In addition, I've added some icons for better usability, which matches to the individual information.

I completely revised the scan menu. For this I have added a small UI in the lower area, which should control the functions.

Since I wanted to make the entire scan process significantly faster, these functions have no further confirmation queries. Which means that the scan is done directly.

With the "Info scan" you will get information like the title or wheather you already own this. Also, the reading status will be displayed.

With "Read" the book is saved directly as "read". Only a confirmation message is shown. The same applies to the "Add" and "Del" functions.

You can also add a whole series with the books symbol button. With this button you just have to scan one book of the series. After successful scanning another menu appears and all volumes of the corresponding series will be shown.

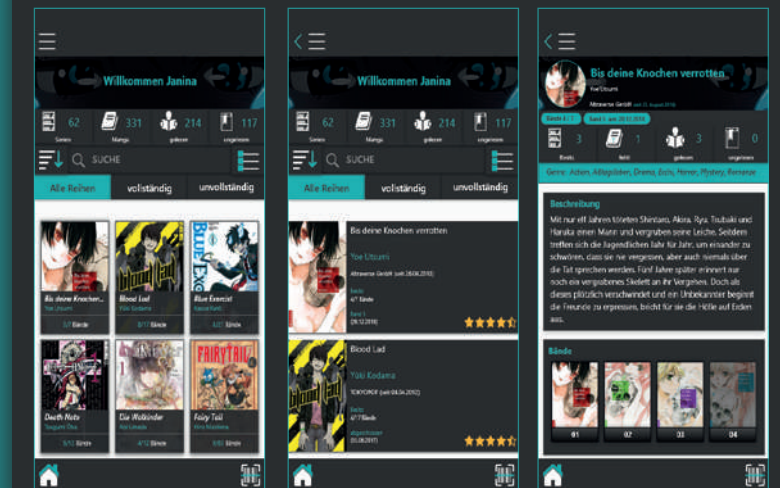
In this menu it is possible to select volumes and either remove or add them.

At this point, it might be useful to include the option to mark the book as read, too.

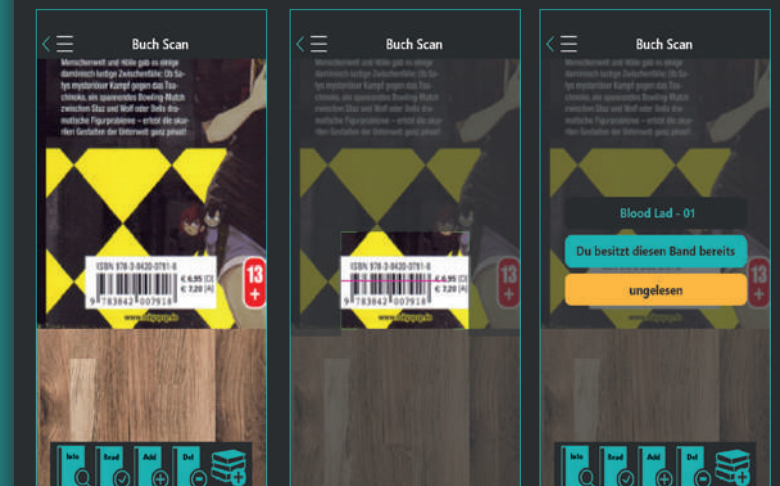
You can also test this prototype here:

<https://bit.ly/2STjvdu>

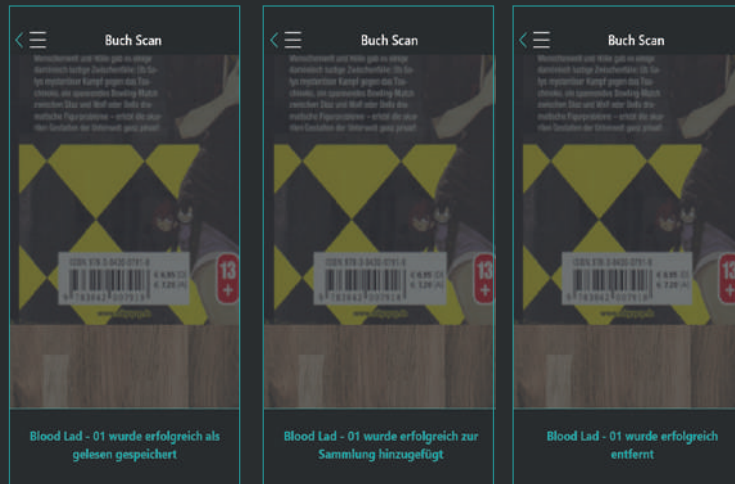
WELCOME & DETAILS



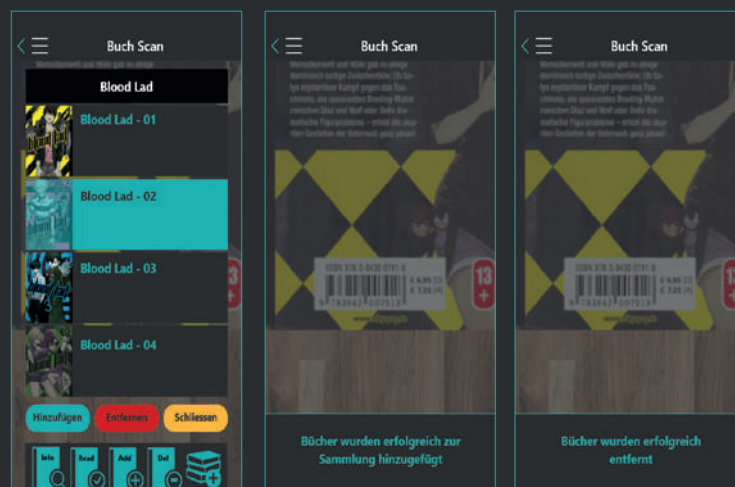
INFO SCAN



READ, ADD & DELETE



ADD VOLUMES



In the second test, I explicitly wanted to know if the app itself is understandable and especially if the informations are clear. Furthermore, I wanted to know if the functions and the structures are user-friendly.

In addition, it asked the people about their general opinion about the app. I wanted to know how well I implemented the app and asked for further suggestions for improvements.

A total of five different users was selected to test this prototype.

Maybe I'll add a little tutorial here. I could also add an option that will give you the choice of which scan menu you would like to choose.

The first scan option should be to scan the book once and then display the advanced options as a menu, as it was the case in the first prototype.

The second scan option could be like it is in the second prototype. So there will be a fast scan option and a normal scan option available.

Another note was that the font size or colors of the design should be freely selectable. Of course these are things that can be added later in an options menu.

Result

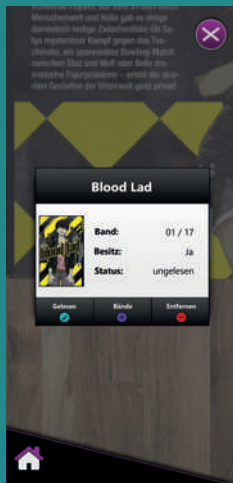
Overall, the users liked the app. All users have understood what the app is for and what benefits the functions should bring.

Some people didn't clearly understand some scanning features. Here I realized that I have to revise this part again in the future.

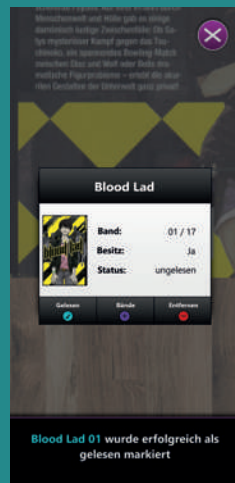
SCAN



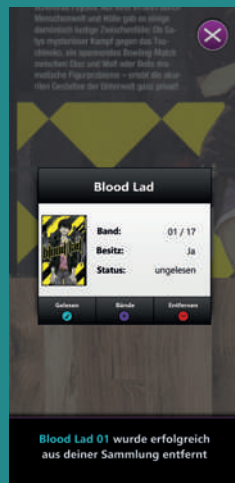
RESULT



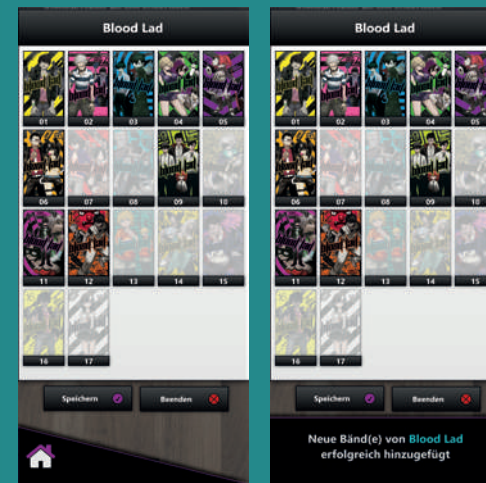
READ



DELETE



ADD VOLUMES



This is the first Hi-Fi prototype I have created with inVision (InVision, 2018). Under the following link you can also test its functionalities:

<https://bit.ly/2Sf0v9a>

The intro should briefly and concisely indicate which main functions the app offers.

In the login area, it should be both the function to create a new account, as well as the resetting of the password.

After successful login you reach the first welcome screen. In this place the already scanned series are displayed.

In addition to a filter function, a search function should also be possible. You can also switch between grid and list view.

In the upper area, there are a couple of statistics for the collected book series and volumes.

If you tap on a book series, you will get to the detail view. Here are some more information about the series. In the lower section you will find the volumes that have been published.

The volumes, which you already own will be highlighted and those you don't have yet are greyed out.

You can tap on the "statistics" on the top, to filter the volumes as "owned", "missing", "unread" or "read".

After a successful scan, a small overview menu is shown. At this place you can also receive some more information about the book.

With the buttons you have the option to mark the book as read, or you can remove it from your list.

The button "Bände +", will open another menu. With this menu you can mark the volumes of the book series, which you own and save them into your list.

Test

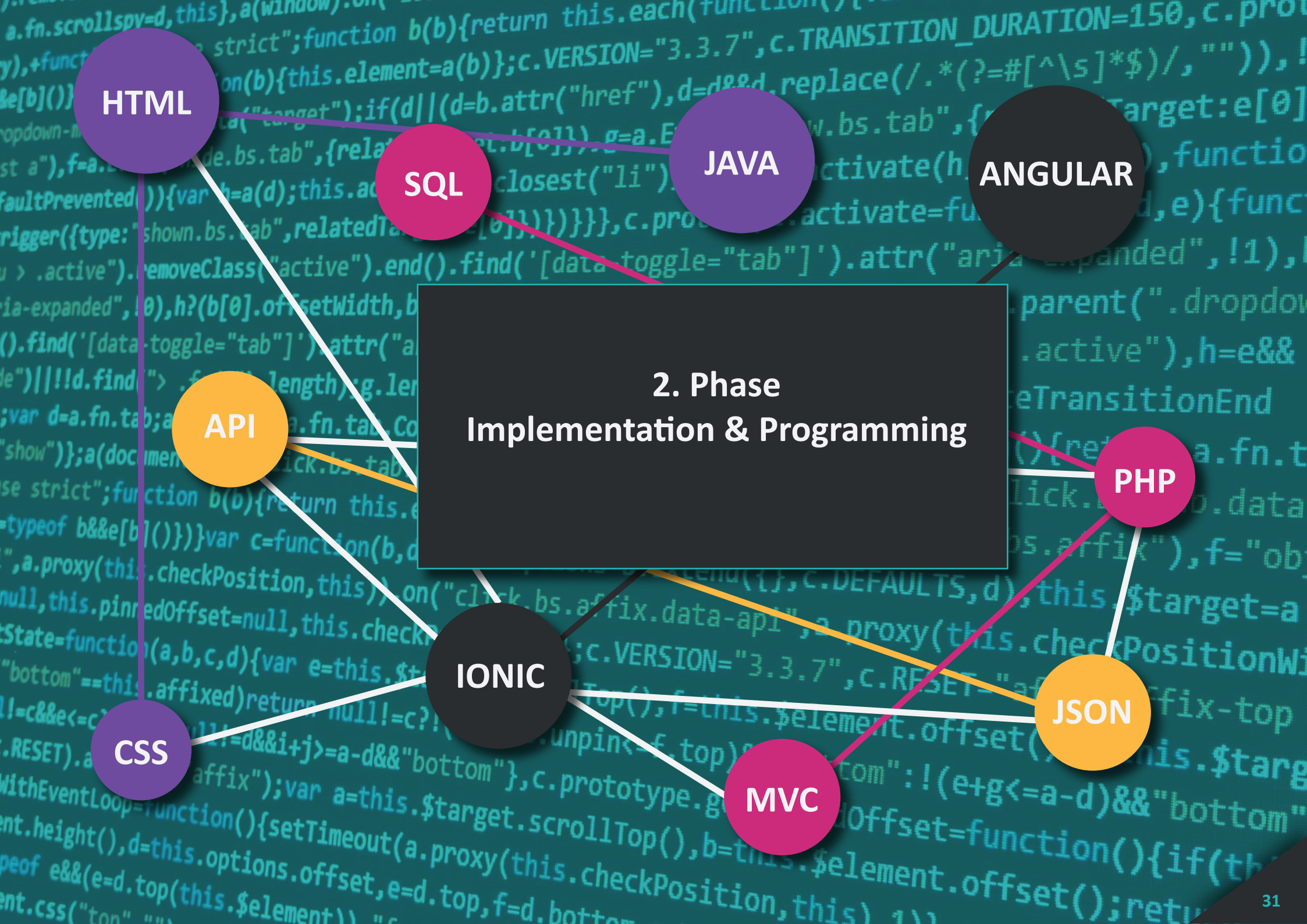
For the test I had a lot of extensive discussions about the design and prototype with my friends, acquaintances and relatives. For this purpose, both the functions were discussed, as well as the information and their arrangements.

Of course, tastes are always different, but I could agree with a few complaints. I have accepted the feedback and suggestions for improvement and included them in the further considerations.

Result

In general, we came to the conclusion, that the design does not look bad. But it reminds on a website and does not fit to a typical mobile app design.

In addition, some pages are overloaded with information that people can't recognize. Also, the scan menu is designed too clumsily.



Expiration of the 2. Phase

I've spent much time thinking about the best way to capture the data I need to realize the app. Especially how to get the necessary information. There are already some APIs on the Internet that can output information about books. Unfortunately, I couldn't find an API, which will transfer exactly the information I need.

So I decided to create my own API in connection with a separate database. I am aware it will be a lot of work to record the individual series and volumes in a separate database. But later there will maybe a possibility to simplify it. However, I can't consider this in the current project due to the little time schedule.

In this phase, I describe how I used to implement the app.

Database

At the beginning, I've designed an ER model with the attributes and relations I need. After the appropriate normalization rules ([Studytonight.com, 2018](https://www.studytonight.com/databases/normalization-rules/)), the relations were split into tables and the attributes in columns, so that a form was created, which no longer receives any redundancies. For the implementing of the relational database, I have used MySQL ([Mysql.com, 2018](https://www.mysql.com/)).

Model View Controller

With a self-created MVC ([Rouse, 2016](https://www.rouse.com/)), I have access to the database using PHP (Php.net, 2018), which will output the information using json ([Json.org, n.d.](https://www.json.org/)). At this point I have also realized a limited system which can capture publishers, authors, series and volumes into the database. This system is only a temporary system. Later, this control panel should be developed properly, so that you can better manage the data.

Security

To have a save system, I've come up with a few things which should prevent to be able to access certain information and data. I could have used systems that had already been developed, but it was significant for me to develop a system for gaining more knowledge in security. Of course, I can't promise 100% security, but through its own implementation, I have learned a lot about it.

API

In order to exchange the data between the programmed app and the database, I have created an API using PHP and json. With PHP, the data is collected from the database and will be presented with json. The APP uses a GET method to access data and display it at the appropriate place. With a POST method in turn, the APP sends data back to the HTTP server and will be transmitted and managed accordingly via PHP.

APP

The APP was programmed with Angular (Angular.io, 2018) and Ionic 4 (beta) (Beta.ionicframework.com, 2018). The main focus was placed on the following pages.

Home

The main page showing the overview of the scanned series and a few statistics.

Details

A page displaying the details of each series, with an overview of each volume and also a few statistics.

Scan

This page manages all scan functions. It includes an information scan that outputs information about the book, a scan function to add volumes and one to remove volumes.

And if you want to add or remove multiple volumes of a series at the same time, you will also find a function. You also have the option to save a book as read. Only the option to save a volume as "unread" again was unconsidered.

App test

In the last step, the programmed app was tested. For this I have previously added the required book series and the corresponding books to the database, so that it was possible to scan these books.

For the realization of the database, I thought about informations I want to capture and store at all. It must of course be possible for users to have their own account. It must capture series, the individual volumes, the associated author and the publisher of the series.

Additional information like genres or target group would not be uninteresting either. Although I considered these in the database planning, they are not included in the current status of the developed app.

The following objects (entities) should be in the database:

Author, Genre, Publisher, Series, TargetGroup, User, Volume

In the next step, I created an ER model with these entities. The ER model is a visual model intended to represent the logical structure of the database. Among other things it should help to develop or implement the relational database. The model should clarify the connection between the individual entities (Singh, 2018).

An ER model consists of the following types:

Entities -> Example: User, Series etc

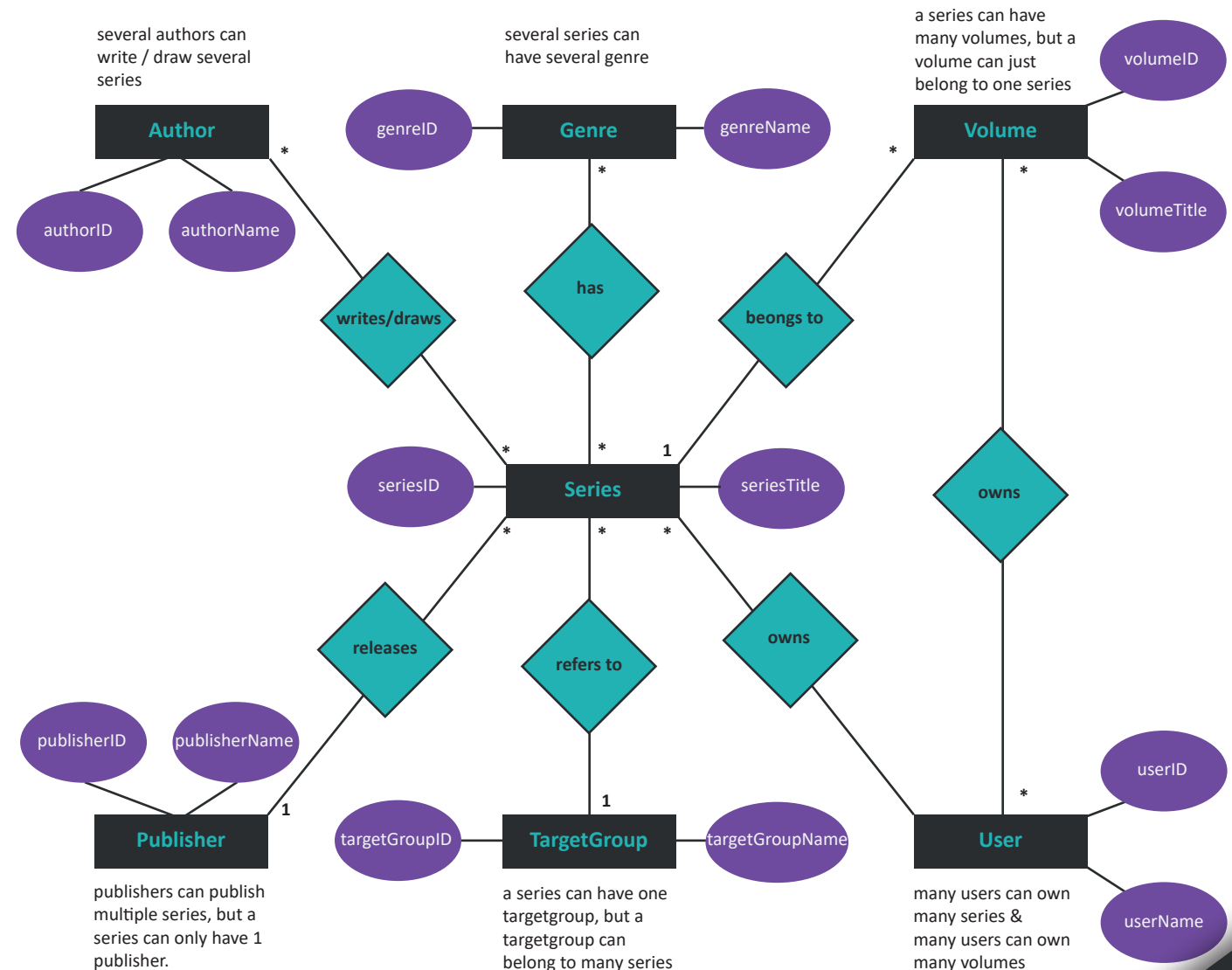
Relationships -> Example: User owns series

Attributes -> Example: Username, UserEmail, etc

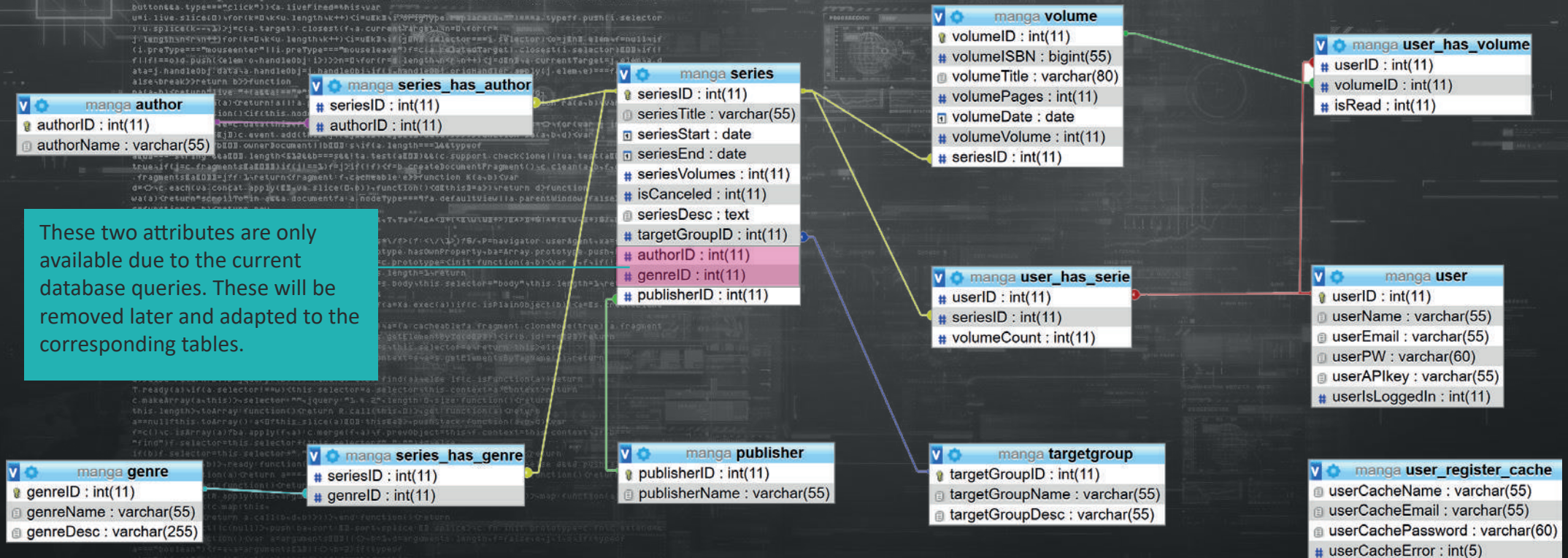
Annotation:

At first I misjudged the relationship between authors and series. I haven't considered that a series can have multiple authors. Unfortunately, I did not recognize it until have I programmed the app itself and added some series and volumes into the database. Same applies to the genre, which, however, I generally have not taken into account in the system until now.

I have adapted these within the database implementation but they are not yet adjusted within the app. Currently the database query can only display just one author per series. However, later I maybe have to reconsider that there is a difference between the draftsman and the author, too. Some book series have both several authors, as well as draftsmen.



Relational Database Model



These two attributes are only available due to the current database queries. These will be removed later and adapted to the corresponding tables.

This table serves only as a "cache" for creating a user. This will be explained in more detail on page.

App development

As mentioned earlier, because of the fact that none of the existing API's that I found provide the information I need for this app, I wrote my own API.

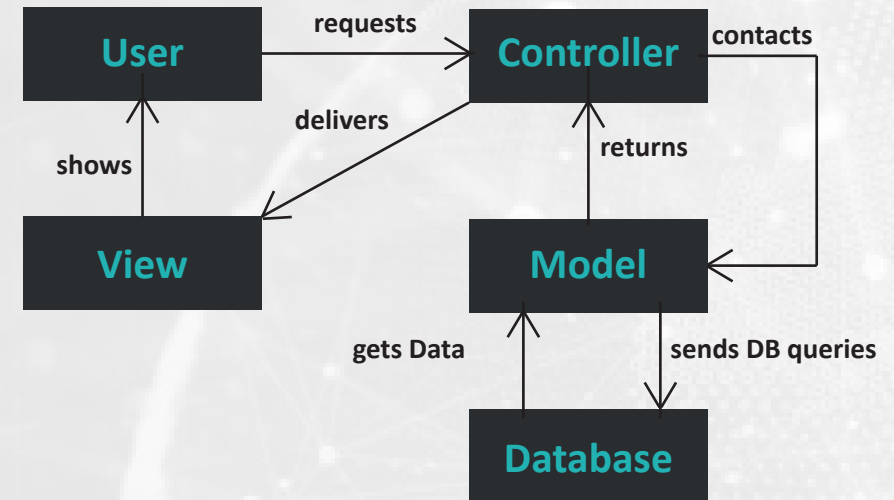
MVC

For this I have used the Model View Controlling system, which I already wrote and used during another project. The MVC separates the code, design and database, allowing a much better overview in the programming phase.

The model files contain the functions for the database queries. These functions are called in the controller files and the queried information are assigned to the view files, which finally display the information in the browser. (in this case the most data will be shown without "view" files)

With a suitable routing system, parameters can be passed within the address line. Thereby methods can be called as well as variables such as user name can be transmitted.

This model briefly describes how a model view controller works



- 1 This part is the root location. The only path, which is accessible
- 2 The name of the controller file
- 3 The method of the controller function, which is called
- 4 The first parameter, which is used to call another method
- 5 A second parameter, which is used as variable \$userName

```
class Access extends Controller
{
    public $user, $userName;
    public $username, $email, $password;

    public function index()
    {
        die("no access!");
    }

    public function user($mode = "", $userName = "")
    {
        $this->userName = $userName;
        if ($mode == "login") {
            $this->login();
        } else if ($mode == "info") {
            $this->info();
        } else if ($mode == "logout") {
            $this->logout();
        } else if ($mode == "register") {
            $this->register();
        } else if ($mode == "response") {
            $this->response();
        }
    }
}
```

(...manga\app\controllers\access.php)

Security

To ensure the security of the data, I have integrated several options.

.htaccess

In order to deny direct access to files, I have created a .htaccess file, with which the appropriate access rights are set and the routing is performed. You will always be redirected to the public folder, and you have no direct access to other folders and files ([Hendrix, 2009](#)).

HTTP headers (CORS)

Since working with an API, also includes working with HTTP headers, I have added some appropriate restrictions (CORS) ([MDN-Web-Dokumentation, 2018](#)). HTTP headers will give you the possibility to use GET and POST requests, which for example will be used, to send form inputs.

But there are also more server requests like DELETE, which, as one might assume, authorize the deletion of data. To increase security and protect the server from unauthorized requests, I have added restrictions to the .htaccess file, so that only POST and GET methods can be executed.

API key

To prevent further direct access to informations, with each login a new API key is assigned the user. This is generated within the APP and transmitted to the server together with the login data.

After successful login, the ApiKey will be added to the user account (in the Database and stored within the APP). Only together with this key, the information can be read from the app.

Password hash

During registration, a hash value is generated from the desired password, so that it is almost impossible to obtain the basic value of the password ([bcryptjs - npm, 2018](#)).

Sanitization

All values, which are transmitted to the server via the app, are filtered with PHP accordingly. This prevents that, by means of script commands, data can be read out or manipulated ([PHP Filters - W3schools.com, 2018](#)).

Prepared statements

To prevent additional SQL injections, all SQL queries are executed via prepared statements. ([PHP Prepared Statements - W3schools.com, 2018](#))

API

With PHP, certain functions are used to collect data from the database. These data are added to corresponding arrays and finally output via json_encode ([JSON PHP - W3schools.com, 2018](#)). If you meet the appropriate access rights, you will then have access to the generated file within the browser.

The programmed app also have access to these generated files and receives the necessary information, which will be displayed accordingly.

In addition, data is sent back to the server using the app, which is managed via the API. For example, a new user is created or the reading status is updated.



Functional explanation

In the following I will explain an example about the functional process of the APP in conjunction with the API

App start

When starting the app, it will be checked, whether the user is already logged in or not. This information will be stored in a local storage and the database. As a result you don't have to log in again and again with every start of the app.

If you are not yet logged in, you will be redirected to the login screen, otherwise to the home screen.

Log in

While logging in, it will be checked whether a username or password has been entered. If this is not the case, an alert screen will be displayed with an appropriate message.

The information about the host is read from the local storage. I've added this to make it easier to change the server. This will then generate the corresponding URL. In addition, an APIkey is generated and set to the corresponding variable.

In the following step, it is assumed that the transmitted data is passed as a string, because if you want to send data to a web server, the data has to be a string

Then the data, username, password and APIkey are sent to the server (url) using the ".post" command and the method this.userInfos() is called.

The userInfo() method will access another URL or rather said another method of the PHP file to get information. This information is generated using SQL queries and presented with Json. With the GET method, the APP has access to the data presented in Json. This data will then be stored in variables.

Subsequently, the variables are sent as parameters to another method, which checks whether the user was able to log in successfully. If this is the case, the information is stored in the local storage and you are redirected to the "home" screen.

If the user couldn't log in successfully, an error message appears.

```
initializeApp() {
  this.platform.ready().then(() => {
    this.statusBar.styleDefault();
    this.splashScreen.hide();
  });
  this.storage.get('userIsLoggedIn').then(userIsLoggedIn => {
    this.userIsLoggedIn = userIsLoggedIn;
    if (!this.userIsLoggedIn) {
      this.router.navigateByUrl('/login');
    } else if (this.userIsLoggedIn) {
      this.router.navigateByUrl('/home');
    }
  });
}
```

\src\app\app.component.ts

```
async login() {
  if (this.user.username === '') {
    this.header = 'Eingabefehler';
    this.msg = 'Bitte Benutzername angeben.';
    this.alert(this.header, this.msg);
  } else if (this.user.password === '') {
    this.header = 'Eingabefehler';
    this.msg = 'Bitte Passwort angeben.';
    this.alert(this.header, this.msg);
  }
  this.storage.get('myHost').then(myHost => {
    this.myHost = myHost;
    this.url = this.myHost + 'manga/public/access/user/login/';
    this.apikey = this.generateAPI();
    const myData = JSON.stringify({
      username: this.user.username,
      password: this.user.password,
      apikey: this.apikey
    });
    this.http.post(this.url, myData).subscribe(
      data => {
        this.data.response = data['_body'];
      },
      err => {
        console.log(err);
      }
    );
    this.presentLoading();
    setTimeout(() => {
      this.userInfos();
    }, 3000);
  });
}
```

\src\app\login\login.page.ts

```
async userInfo() {
  this.storage.get('myHost').then(myHost => {
    this.myHost = myHost;
    this.url =
      this.myHost + 'manga/public/access/user/info/' + this.user.username;
    this.userApi = this.apikey;
    this.http
      .get(this.url)
      .map(res => res.json())
      .subscribe(
        data => {
          this.userID = data.user[0]['userID'];
          this.userName = data.user[0]['userName'];
          this.userIsLoggedIn = data.user[0]['userIsLoggedIn'];
          this.check(
            this.userID,
            this.userName,
            this.userApi,
            this.userIsLoggedIn
          );
        },
        err => {
          console.log(err);
        }
      );
  });
}
```

\src\app\login\login.page.ts

```
async check(userID, userName, userApi, userIsLoggedIn) {
  if (userIsLoggedIn === '1') {
    this.storage.set('userIsLoggedIn', true);
    this.storage.set('userID', userID);
    this.storage.set('userName', userName);
    this.storage.set('userApi', userApi);
    this.router.navigateByUrl('/home');
  } else {
    this.header = 'Login fehlgeschlagen';
    this.msg = 'Bitte überprüfe deine Eingabe.';
    this.alert(this.header, this.msg);
  }
}
```

\src\app\login\login.page.ts

```
generateAPI() {
  this.str = '0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890';
  this.strPos = 0;
  for (let i = 0; i < 32; i++) {
    this.block =
      Math.random()
        .toString(36)
        .substring(2, 15) +
      Math.random()
        .toString(36)
        .substring(2, 15);
    const chunks = this.block.match(/.{1,6}/g);
    this.block = chunks.join('-');
    return this.block;
  }
}
```

\src\app\login\login.page.ts

```
public function login()
{
    $postData = file_get_contents("php://input");
    if (isset($postData)) {
        $request = json_decode($postData);
        $username = filter_var($request->username, FILTER_SANITIZE_STRING);
        $password = filter_var($request->password, FILTER_SANITIZE_STRING);
        $apikey = filter_var($request->apikey, FILTER_SANITIZE_SPECIAL_CHARS);
        $user = $this->model('User')->getUser($username);
        if ($password_verify($password, $user->password)) {
            $this->model('User')->update($user->userID, $apikey, 1);
            $this->model('User')->update($user->userID, $apikey, 1);
        }
    }
}
```

\manga\app\controllers\access.php

```
public function info()
{
    $items_arr = array();
    $items_arr['user'] = array();
    $this->user = $this->model('User')->getUser($this->user->username);
    if ($this->user['userIsLoggedIn'] == 0) {
        $user_item = array(
            "userID" => "",
            "userName" => "",
            "userAPIkey" => "",
            "userIsLoggedIn" => $this->user['userIsLoggedIn'],
        );
        array_push($items_arr['user'], $user_item);
    } else {
        $user_item = array(
            "userID" => $this->user['userID'],
            "userName" => $this->user['userName'],
            "userIsLoggedIn" => $this->user['userIsLoggedIn'],
        );
        array_push($items_arr['user'], $user_item);
    }
    echo json_encode($items_arr);
}
```

\manga\app\controllers\access.php

```
// 201812211220722
// http://needle-sorcery.com/manga/public/access/user/info/janina

{
  "user": {
    {
      "userID": "1",
      "userName": "Janina",
      "userIsLoggedIn": "1"
    }
  }
}
```

```
\manga\app\models\User.php

public function setLoggedIn($userID, $api, $userIsLoggedIn)
{
    $this->db->connect();
    $this->sql = "UPDATE user
    SET userIsLoggedIn = :userIsLoggedIn,
    userAPIkey = :api
    WHERE userID = :userID";
    $this->db->query($this->sql);
    $this->db->bind(':userID', $userID);
    $this->db->bind(':api', $api);
    $this->db->bind(':userIsLoggedIn', $userIsLoggedIn);
    $this->db->execute();
    $this->db->close();
}
```

```
public function getUser($userName)
{
    $this->db->connect();
    $this->sql =
        "SELECT * FROM user
        WHERE userName = :userName";
    $this->db->query($this->sql);
    $this->db->bind(':userName', $userName);
    $this->db->close();
    return $this->db->result();
}
```

User Test

For testing of the programmed app, two important things had to be considered first. The app was initially programmed only for android. Therefore, the tester should own an android device. The second significant precondition is the tester owns the series that have already been entered into the database.

Because of this restriction, only one user test was possible for now. More tests will follow as more series are entered into the database. Then the probability will be higher to find testers who own the added series.

The test procedure was as follows.

The tester should independently explore the entire app. My task was initially only to observe. Then I asked a few questions.

Observation:

The tester has installed the app and created a new account. With this account, the tester has subsequently logged in successfully.

Of course, since the user account is new, there was still no information on the home screen. In this way, the tester first considered the scan button, which he directly recognized. First, the individual functions were tested in the Scan UI. It turned out that sometimes it can come to problems with the detection, especially if books are scanned too fast in a row. The tester quickly understood and extensively tested the individual scan functions.

After scanning, the tester navigated back to the main screen and initially looked at the statistics. The structuring and Information of the shown series have been recognized.

Next, the tester "clicked" on a certain series and were directed in the corresponding detail view. The tester looked around extensively and tried with some more tabs on the screen to get more information, which are unfortunately not yet available.

With a tab on the lower home button, he quickly returned to the main screen.

Result:

After extensive observation and questioning, I came to the realization that the scan functions are well understandable, but only after extensive trial and error. Here I should definitely add an explanation. On some devices, the font size seems to be very small, which should definitely be adapted. The tester felt very positive about the general structuring and design. Only the scan menu should get an improvement.

Of course, a few wishes and suggestions were also mentioned, which should still include the app in the future. Such as a general overview of the series. At the moment you only have the possibility to get details about the series that you have scanned. This tester has also mentioned that he would definitely use this app, as he has not found an similar app with such functions and informations until now.



USER GUIDE

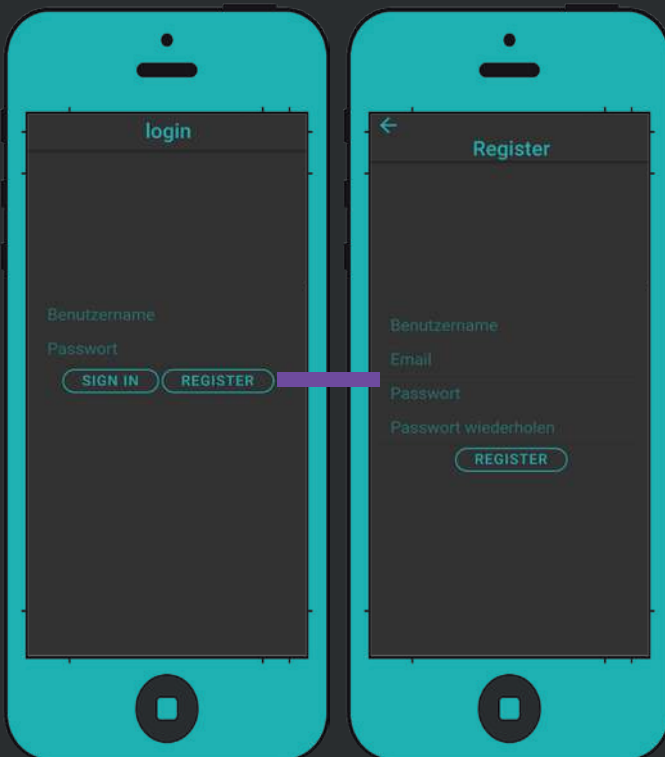


How to use

Manga Collection Royale

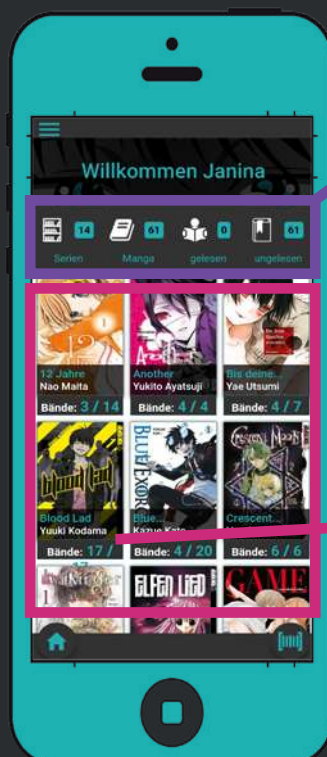
Version 1.0.0.1





The login screen will take you to the registration screen by using the "Register" button. After successful registration, you can navigate back to the login screen and log in with your data.

Both during registration and when logging in a response to input problems with corresponding error messages will appear.



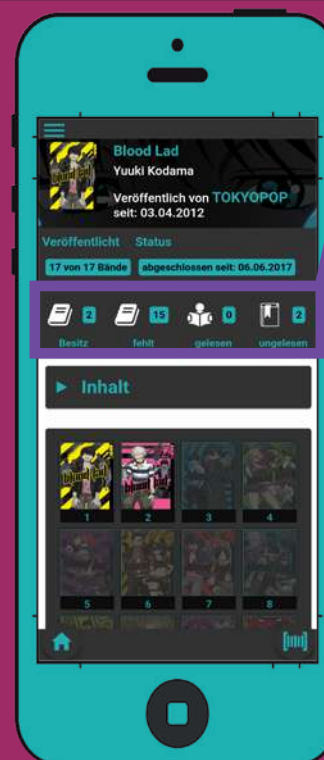
In the center of the welcome screen, all series which you have already scanned and added will be displayed.

A small Coverimage (the cover of the first volume of the series) is displayed. Among them a short form of the title and the main author of the series will be shown.

Also, there is a small number, which shows the amount of volumes that you already own and how many there are in total. When tapping on one of the series, you'll get to a detailed view.

At the top of the welcome screen you'll find the total statistics, all the saved books and series of your own collection. (Series, Manga, read, unread)

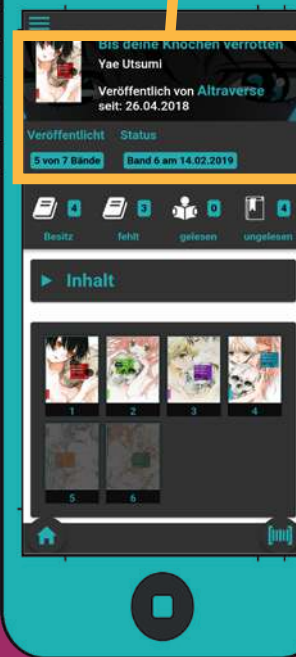
The detailed view also contains statistics. These, however, relate to the selected series. Thus you'll receive information, how many volumes of this series you already own, how many you're still missing and how many of them are read or unread.

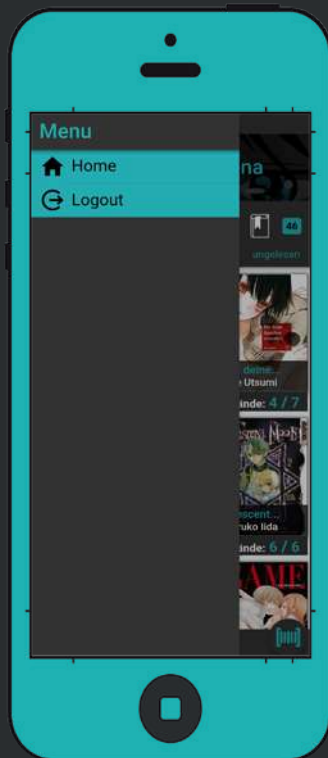


The detailed view gives you more information about the desired series. Here you also get a small text about the content of the series, which you can open and close.

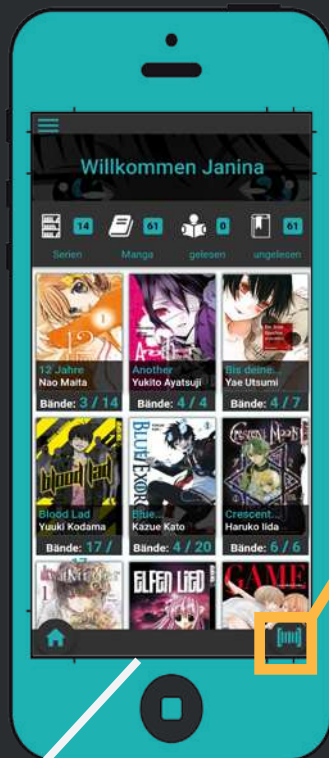
Below is a view of the volumes belonging to the series. Color highlighted volumes are the ones you already own. All grayed out are missing in the collection. Note that volumes that have not yet been published, but whose date of publication is already known, are also displayed in this list, which of course are also missing.

Above the statistics are the general information about the series such as Title, Author or Publisher. In addition, there is information about the current status of the already published volumes, as well as the status, whether it's already completed or when the next volume appears.





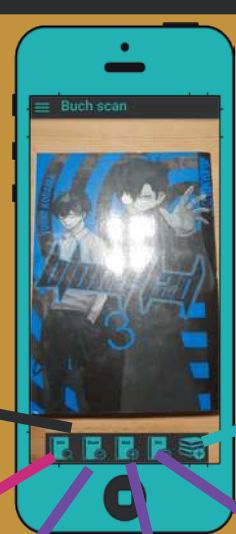
With the menu, it is currently only possible to switch back to the welcome (Home) screen, or to log out.



There is a menu at the bottom of the welcome screen and on the details page. The house icon navigates, as you can imagine, to the main screen. With the "Scan" button on the right side, it is possible to navigate to the scan page.



With each of the lower buttons of the Scan UI, the scan function is initially started, which makes it possible to capture ISBN codes.



With the info scan, you get information about whether you already own a volume and also, if you have already read it or not.

This is especially useful when buying new books and you are not sure, if you already have this volume in your collection.

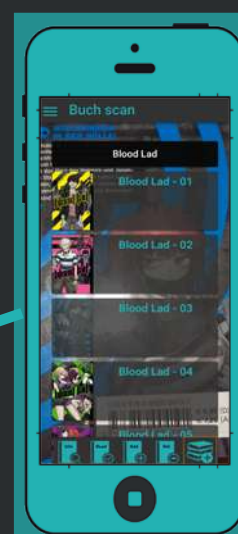


With the "Read" scan function, the book will be updated as read, if you have already saved the book in your collection. Otherwise a message appears, which will tell you, that this book isn't in your collection yet. If saved successfully, a message will appear at the bottom of the screen.



The next scan function "Add" is used to add a book. If it is the first book in a series, the series (not the corresponding volumes) is also saved directly into the collection.

After success, a message will be shown. However, if you already own the book, a error message is displayed.



With the last function in the UI, it is possible to add or remove a complete book series. All you have to do is scan one book of the series and you get a complete list of all the volumes belonging to it.

Here you can select all volumes and either remove or add them all. Books that you already own are highlighted by the cover, and those you do not own are greyed out

Of course there is also a function to remove books - the "Del" function. With this function, the books are removed from the list.

CONCLUSION

I started the project with the idea that manga collectors might have the problem of losing track of their collection. So I wanted to find out if this problem really exists and how manga collectors deal with the problem. What options exist to eliminate the problem?

Since there is an APP for nearly every possible problem these days, I wanted to know if people are using or want to use an app for this potential problem as well. Of course, I was previously aware that there are already APPs with which it is possible to scan books, but what are their advantages and disadvantages?

With a targeted internet research, I could find more and more people with the same problem. Even with a survey, I could find out that it can become a problem the bigger the collection is. So I've looked more closely at the existing APPs to see why these are not so popular for manga collectors.

The problem of these APPs are, that scanning takes far too long. Sometimes it takes up to four "clicks" to finally save a book. This is especially annoying if you have large collections.

Also, it is particularly inconvenient if you own manga series, with a high number of volumes. Having to scan every single book in a book series probably will also scare off many collectors from managing their collections with an APP.

That's why I wanted to find out which information are especially important for manga collectors. Most collectors like to have statistics. For example, how many books they own, and most of all, how many are read or unread.

Another important piece of information is knowing when the next book of a series will be released. Of course, some other important things were also mentioned, but these would have been impossible to include within the short time.

I tried to adapt the design to the theme as much as possible and clearly display the most important information. It should be distanced a bit of the standard design, but still be clearly recognizable as an APP.

With some user testing, I found out that some people might still have some problems with some parts of the APP.

Especially the scan menu is not yet clear for everyone.

The programming itself was probably the most interesting part. Because implementing all of your ideas the way you imagine them is not always easy.

Nevertheless, I could nearly implement everything as I planned before. With the CameraPreview (Camera Preview - Ionic Framework, 2018) function and the displayed menus, I wanted to bring a bit of augmented reality similarity to the APP.

I also wanted to try to program my own secure log-in system. It was important for me to fully engage with it and to develop my own way to make the system as safe as possible. For this I have developed a function that assigns a new APIKey to the user with each login. Only with this key it is possible to access the corresponding data.

I've also built a fast and simple interface that lets you store publishers, authors, manga series and manga volumes in the database. In doing so, I have integrated small but maybe not entirely uninteresting features.

Each book has its own cover, which should be displayed. These must of course be present at the appropriate place. However, it is very cumbersome to create each image with the matching image title and then upload it.

Therefore, I've developed a small feature, which automatically generates placeholder images with the corresponding titles, according to the number of books in the series.

All in all, I've developed an app that allows users to register and log in with their own account. You have an overview of your already scanned books and a small detail view of the individual series.

You have the option of performing an information scan that indicates whether you already own the book or not. It is possible to mark a book as a read, add or remove it. If you want to add or remove an entire series of books, you have the opportunity to do so.

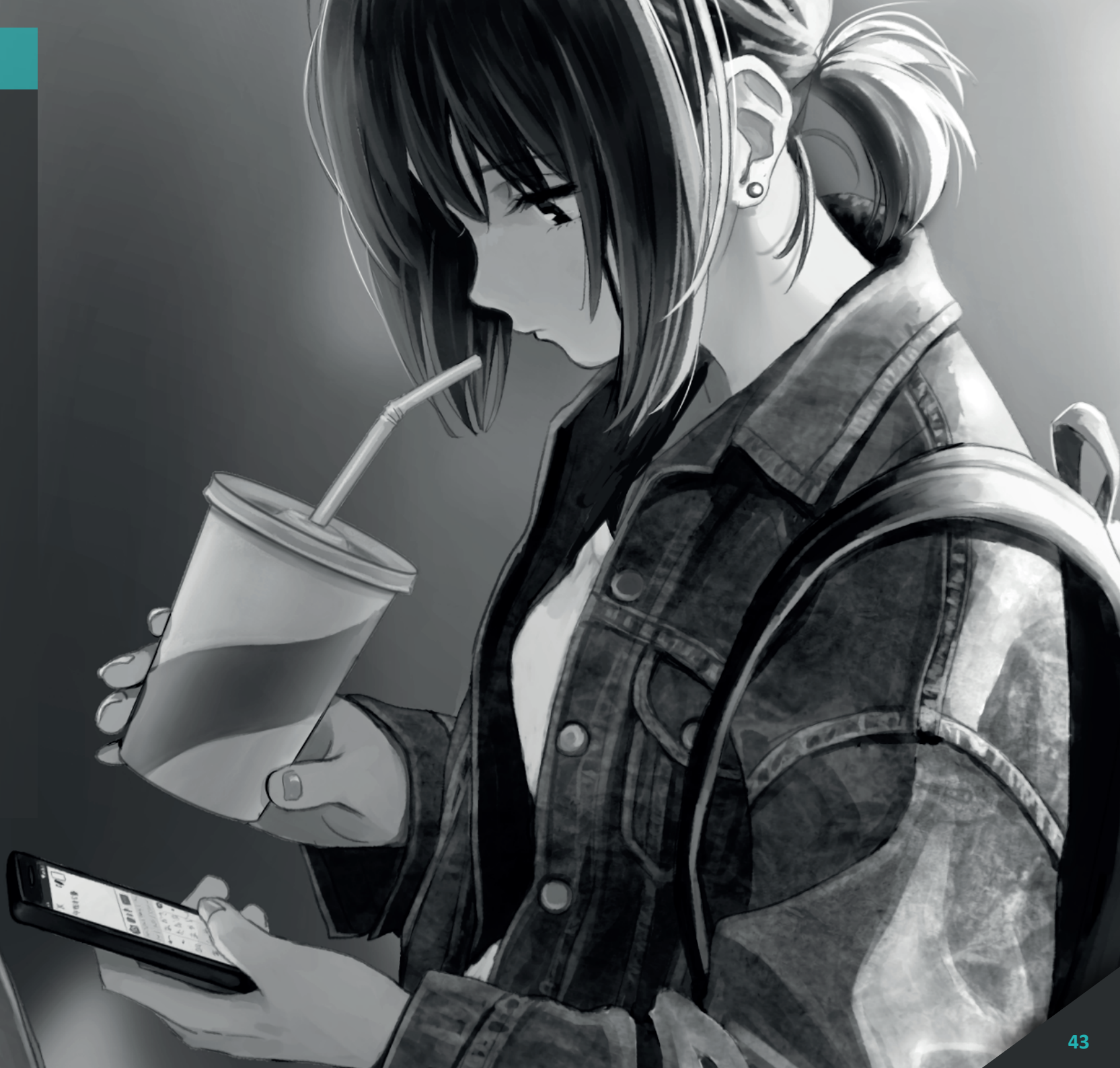
Sometimes it was a technical and mental challenge, but I will still continue to work on this app in the future and try to adapt it even more specifically.

PERSPECTIVES

In my opinion, this APP can be used by many manga collectors in the future, if it will still be adjusted to the user. Currently there isn't an APP available which is specifically designed for manga collector. Most functions of existing APPs are too cumbersome for users. Above all, the function which indicates when you already own a book, appeals very well by users.

In the future I will pay particular more attention to the fact that the scanning system will be more understandable for everyone. I will add a little tutorial that will explain the scan system. Furthermore, I will add the possibility to choose between two different Scan menus.

Generally, there are many things that can and should be added to the system. Like, for example, a general overview of the series. After all, many people also want to have information about book series which they don't own.



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Appendix

Prototype Tester Nr. 1

1. Is it obvious to you what this app is for?

Yes

2. How do you like the overall design of the app?

The design for a prototype would definitely be very solid, but would have some design suggestions for the post-prototype phase.

3. What information did you collect?

Collected Manga Series with Volume Number (Owned and Total), Mangaka, Publisher, My Personal Collection Stats, Synopsis and all volumes of the series

4. Was the information clearly visible to you?

Yes

5. What do you think about the structuring of the app?

The structuring is by itself User-friendly. There are only a few things I would like to be added.

6. Were the functions of the individual scanning options understandable for you?

Except for the info button I knew at first glance what each option serves. But for the info Button I could also think what this could be for.

7. What do you think about the implementation of the scan functions in general?

Personally, I would integrate scanning with no options, adding only volumes and putting the rest of the options in the series as a menu.

8. Your general opinion about the app, which improvements would you suggest?

Definitely a cool idea and with some adjustments a great app. I have already mentioned a suggestion in point 7. Otherwise I would add a sorting after "% completed" of the series

Appendix

Prototype Tester Nr. 2

1. Is it obvious to you what this app is for?

Yes, to capture what manga you already have in your collection and how far you are in your reading progress.

2. How do you like the overall design of the app?

In some places somewhat confusing, especially in the scan function.

3. What information did you collect?

You can capture which manga you have and these are divided into series and volumes. There is also a short summary for each series.

4. Was the information clearly visible to you?

It was clear and easy to see how many volumes you have of each series.

5. What do you think about the structuring of the app?

6. Were the functions of the individual scanning options understandable for you?

The last feature with several mangas is not really obvious to me. Is that to scan several manga at the same time? The rest was very understandable.

7. What do you think about the implementation of the scan functions in general?

Without live demo or test difficult to answer. To add basically good, but to mark as "read" or remove rather suboptimal.

8. Your general opinion about the app, which improvements would you suggest?

Generally good idea, in the implementation in the demo, however, a few shortcomings.

Appendix

Prototype Tester Nr. 3

1. Is it obvious to you what this app is for?

Yes, the app is self-explanatory.

2. How do you like the overall design of the app?

The design is simple but you should make the scan option more visible for a release version.

3. What information did you collect?

You could capture a navigation through collected tapes, scanning and adding books and marking as read and unread.

4. Was the information clearly visible to you?

Clear and noticeable,

5. What do you think about the structuring of the app?

Good, but I would welcome adaptation of colors and text size.

6. Were the functions of the individual scanning options understandable for you?

Yes

7. What do you think about the implementation of the scan functions in general?

Very good, but with Info perhaps an option for the short description of the book would be a nice optional feature. But apart from that, the individual options are easy to recognize and easy to use.

8. Your general opinion about the app, which improvements would you suggest?

I think the app has great potential and would certainly make one or the other Mangafan happy. Improvements would only be in the sense of personal adaptation of, for example, colors.

Appendix

Prototype Tester Nr. 4

1. Is it obvious to you what this app is for?

Yes

2. How do you like the overall design of the app?

It's OK

3. What information did you collect?

4. Was the information clearly visible to you?

Yes

5. What do you think about the structuring of the app?

Not necessarily self-explanatory, but rather try until you know what you have to do.

6. Were the functions of the individual scanning options understandable for you?

Preferring to show options after detecting the barcode than before.

7. What do you think about the implementation of the scan functions in general?

8. Your general opinion about the app, which improvements would you suggest?

Seeing when the next volume will be published would be the most important information for me.

Appendix

Prototype Tester Nr. 5

1. Is it obvious to you what this app is for?

Yes, it is actually obvious and immediately recognizable.

2. How do you like the overall design of the app?

It is well structured, only the fonts could partially better be adjusted.

3. What information did you collect?

I could see a collection of manga, how many volumes of the series are already collected. Some statistics. And further details about the individual series.

4. Was the information clearly visible to you?

Yes

5. What do you think about the structuring of the app?

6. Were the functions of the individual scanning options understandable for you?

Although you can generally understand for what the individual buttons should be, but an explanation might be useful.

7. What do you think about the implementation of the scan functions in general?

The implementation is good but it has to be optimized a bit.

8. Your general opinion about the app, which improvements would you suggest?
